BALANCING REPRESENTATIONS BY IDENTIFYING AND GENERATING UNDERREPRESENTED DATA

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ABSTRACT

Self-supervised learning inherits gaps in its source data. We study whether these gaps can be repaired while representation learning is in progress. We introduce BRIDGE, a label-free feedback loop in which the current SSL representation scores local support, a budgeted acquisition policy selects sparse but recoverable regions, and a frozen generator adds local variants before continued pretraining. The resulting intervention changes the representation used for later acquisition. Across five long-tailed and web-source regimes, BRIDGE improves average linear-probe and kNN transfer over source-matched SSL baselines. Paired full-schedule experiments further show gains with both ResNet-50 and ViT-S, and on four of five SSL objectives spanning contrastive, self-distillation and masked-reconstruction pretraining, while low-label fine-tuning and semantic segmentation extend the evidence beyond frozen-feature evaluation. Matched controls compare adaptive repair with one-shot and frozen policies, conventional augmentation, VLM-guided text-to-image generation, and an AIDE-style acquisition pipeline. We formalize the acquisition problem through recoverability-weighted marginal support utility. The analysis gives conditions for an interior repair region and explains why allocation priorities can change after an intervention. Isolating the loop from its components gives the clearest result: after a single repair stage, spending the budget on the densest quartile of the representation and on its extreme tail are indistinguishable, with a spread of 1.32 points across the entire score range, while the same acquisition rules separate by 8.0 points once the signal is recomputed over five stages. The benefit of targeted acquisition is therefore produced by the feedback loop and not by any individual intervention it makes.

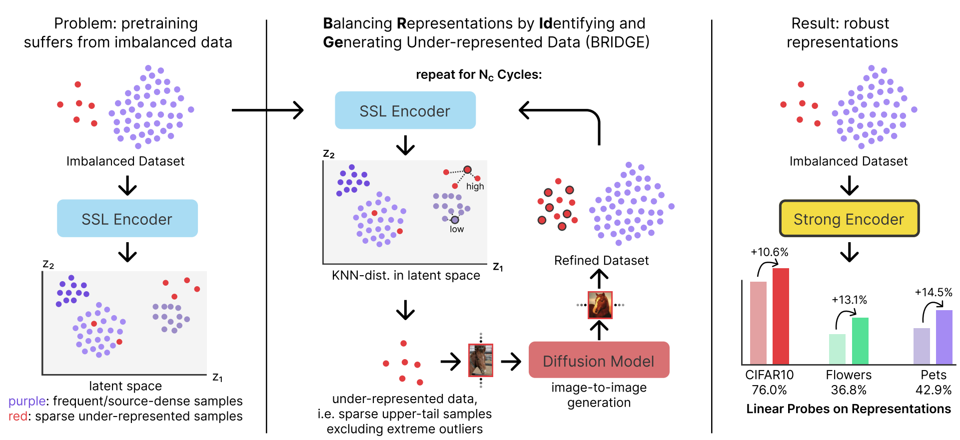

Figure 1: Conceptual overview of BRIDGE. We alternate between SSL pretraining, sparsity scoring in representation space, and targeted image-to-image diffusion augmentation. BRIDGE augments a mode-centered band inside the upper OOD tail and not the extreme tail itself.

1 INTRODUCTION

Self-supervised learning is now a standard way to learn transferable visual representations from unlabeled images. Strong methods span contrastive learning, masked prediction, and self-distillation (Chen et al., 2020; He et al., 2020; 2022; Caron et al., 2021; Oquab et al., 2024; Siméoni et al., 2025). Yet these methods still inherit coverage biases from the source corpus. Prior work shows that imbalance can hurt self-supervised learning and motivates specialized long-tailed training strategies (Tian et al., 2021; Assran et al., 2023; Kukleva et al., 2023; Jiang et al., 2021). Learned features can concentrate around frequent modes while providing weaker coverage of underrepresented regions. The experiments in Section 6 test whether adding data to such regions improves transfer.

Most prior attempts to address long-tailed SSL are model-centric. Temperature Schedules reshapes the optimization trajectory of contrastive learning over time (Kukleva et al., 2023). SDCLR introduces a self-competitor to emphasize harder and rarer samples (Jiang et al., 2021). These methods are strong baselines, but they still treat the source distribution as largely fixed.

We study the complementary direction and ask whether we can repair the source data while SSL training proceeds. We propose BRIDGE, a cyclic data-repair loop shown in Figure 1. After each pretraining stage, we embed the current training set, score each image by mean kNN distance, select a sparse but still semantically recoverable region of representation space, and generate new training images for that region with a frozen diffusion model. The next cycle then trains on the repaired dataset. The encoder reveals where the source manifold is thin, and the generator densifies those regions before the next round of learning.

Two aspects of BRIDGE are central. First, BRIDGE restricts attention to the upper sparse band of the OOD distribution, finds the mode within that band, and then selects a diverse set of anchors around that mode. This avoids the super-OOD tail, which often contains noisy or weakly useful images, while retaining the underrepresented region that is most amenable to repair. In the ablations, we compare this selector against top-tail and uniform alternatives. Second, we evaluate BRIDGE across five distinct source regimes. In addition to ImageNet-100-LT, CIFAR-10-LT, and CIFAR-100-LT, we study web-source unlabeled data drawn from PASS and DiffusionDB. Across these five regimes, BRIDGE is best on the seven-task average and remains strong under both linear-probe and kNN transfer.

We make three contributions. First, we formulate data repair as recoverability-weighted support allocation and derive conditions under which a finite budget should favor sparse interior regions and update its allocation after repair. BRIDGE gives this formulation a label-free cyclic implementation. Second, we introduce an upper-tail mode-window rule with diversity-aware subsampling, which targets semantically recoverable sparse regions without drifting into the extreme OOD tail or wasting budget on near-duplicates. Third, we evaluate five source regimes and seven transfer tasks, with paired ViT-S experiments, full fine-tuning, semantic segmentation, and matched controls for alternative feedback and generation pipelines. Fourth, we separate the acquisition rule from the loop that applies it by running the same rules under one repair stage and under five. The rules are indistinguishable under one and separate by 8.0 points under five, which locates the contribution in the repeated re-estimation and not in the selection heuristic considered on its own.

2 BACKGROUND AND RELATED WORK

Skewed SSL and representation-driven data selection. Long-tailed SSL methods modify training while leaving the source set fixed. SDCLR emphasizes persistently difficult samples, Temperature Schedules changes contrastive optimization, and COLT introduces external OOD data (Jiang et al., 2021; Kukleva et al., 2023; Bai et al., 2023). Other studies examine hidden imbalance, frequency-aware learning, and uncurated data (Tian et al., 2021; Assran et al., 2023; Lin et al., 2023). Data curation uses representations to retain informative subsets or remove redundancy (Whang et al., 2023; Sorscher et al., 2022; Abbas et al., 2024). Earlier augmentation methods instead add support. SMOTE interpolates labeled minority examples, while DOPING oversamples low-density latent regions for unsupervised anomaly detection (Chawla et al., 2002; Lim et al., 2018). BRIDGE uses the current SSL representation to decide where new source images should be added, then recomputes that decision after learning from them.

Budgeted acquisition and coverage. Active learning studies the allocation of a finite acquisition budget and supplies the closest formal precedent for our selection rule. The core-set formulation treats acquisition as k-center coverage of the representation and selects with the same greedy farthest-first traversal we use, under the approximation guarantee of Gonzalez (1985), though it acquires labels for existing images and does not create new ones (Sener and Savarese, 2018). ProbCover reformulates the objective as maximizing covered probability mass at a fixed ball radius (Yehuda et al., 2022). Most relevant to our results, Hacohen et al. (2022) show that the best acquisition strategy reverses with budget: dense and typical regions are preferable when the budget is small relative to the corpus, and sparse or uncertain regions become preferable only once it is large. Our single-stage sweep in Section 6.4 is consistent with that reversal, since one repair stage leaves every score band indistinguishable, and the advantage of the sparse band appears only as repeated stages accumulate. Two differences separate our setting from this literature. Acquisition here adds generated images and consumes no labels, so the budget is limited by generation cost and not by annotation, and the scoring model is the SSL encoder being trained, so each acquisition changes the function that selects the next one.

Generative augmentation. Diffusion models make image-conditioned repair possible through controlled denoising (Ho et al., 2020; Rombach et al., 2022; Meng et al., 2022). DA-Fusion applies this mechanism to few-shot supervised recognition, and TADA targets examples learned slowly under a supervised objective (Trabucco et al., 2024; Nguyen et al., 2026). Larger studies show that synthetic-data utility depends on prompt design, filtering, diversity, generator configuration, and domain gap (Azizi et al., 2023; He et al., 2023; Fan et al., 2024). BRIDGE uses neither class identity nor supervised learning speed. It allocates a fixed generation budget from local support in an unlabeled representation.

Synthetic representation learning and data engines. Synthetic imagery can serve as the full training source or define contrastive views (Ren and Lee, 2018; Baradad et al., 2021; 2022; Jahanian et al., 2022; Sariyildiz et al., 2023; Tian et al., 2023). DiffAug is the closest iterative SSL method. It jointly learns a conditional generator for positive views, while BRIDGE holds the generator fixed and studies where it should intervene in an observed source corpus (Zang et al., 2024). AIDE and industrial data engines also close a model–data loop, but identify supervised task failures and use language models, annotation, and domain-specific verification (Liang et al., 2024; Worker and Pathak, 2022). We evaluate language-mediated acquisition and text-to-image generation as direct alternatives.

External representation priors. DreamTeacher distills generative features into discriminative backbones, DIFT extracts semantic diffusion features, and REPA transfers visual features in the opposite direction into a diffusion model (Li et al., 2023; Tang et al., 2023; Yu et al., 2025). A frozen generator therefore contributes knowledge from external pretraining even when its parameters never change. Our controls compare direct feature transfer, uniform generation, and targeted generation from the same model. The remaining question is how an unlabeled learner should allocate a limited intervention budget as its representation evolves. Section 3 formulates this problem.

3 Theoretical Framework

We formulate targeted data repair as a budgeted allocation problem over coherent regions of representation space. A region is valuable when additional support can reduce its contribution to downstream risk, while a repair is useful only when the generator preserves the semantic content of its anchor. The acquisition policy must therefore balance support deficit, repairability, and redundancy.

Let $r_k(z; Z)$ be the distance from z to its kth neighbor in representation set Z. Let A be a set of valid generated variants associated with z.

Proposition 1 (Local support monotonicity). *For any A, $r_k(z; Z \cup A) \leq r_k(z; Z)$. The inequality is strict whenever an added point enters the first k neighbors and displaces a point at strictly larger distance.*

The statement follows because enlarging the candidate set cannot increase its kth order statistic. To connect support to learning, partition representation space into coherent regions j with current effective support n_j. We model a regional learning curve by

$$\mathcal{B}(\mathbf{m}) = \sum_j p_j a_j (n_j + m_j + \kappa)^{-\gamma}, \tag{1}$$

where p_j is the transfer-relevant mass of region j, a_j is its difficulty, m_j is the number of valid repairs allocated to it, $\kappa > 0$, and $\gamma > 0$. The marginal reduction from one valid repair is

$$\Delta_j(m_j) = p_j a_j \left[(n_j + m_j + \kappa)^{-\gamma} - (n_j + m_j + 1 + \kappa)^{-\gamma}\right]. \tag{2}$$

Proposition 2 (Sparse-region marginal value)**.** *$\Delta_j(m_j)$ is positive and strictly decreases with $n_j + m_j$. When two regions have equal transfer mass and difficulty, the region with less effective support has the larger marginal value.*

This follows from strict monotonicity and convexity of $x \mapsto x^{-\gamma}$. It provides a direct reason to allocate a limited budget away from regions that are already well supported.

Generation can fail, especially for corrupt or semantically incoherent anchors. Write d for a sparsity score, $b(d)$ for marginal value conditional on valid repair, $\rho(d)$ for the probability of a faithful and useful variant, and $\lambda \geq 0$ for the cost of an invalid variant. Expected repair utility is

$$u(d) = \rho(d)b(d) - \big(1 - \rho(d)\big)\lambda. \tag{3}$$

Proposition 3 (Interior recoverable-support optimum)**.** *Suppose b and ρ are positive and differentiable on $[d_{\min}, d_{\max}]$, with $b'(d) > 0$ and $\rho'(d) < 0$. Define*

$$g(d) = \frac{b'(d)}{b(d) + \lambda} + \frac{\rho'(d)}{\rho(d)}. \tag{4}$$

If g is strictly decreasing, $g(d_{\min}) > 0$, and $g(d_{\max}) < 0$, then u has a unique maximizer $d^\star$ in the interior, characterized by $g(d^\star) = 0$.

The condition compares the relative growth of support value with the relative decay of repairability. Dense regions are preferred while value grows faster, and utility begins to fall once reliability decays faster. The maximizing score depends on the source distribution and generator through b and ρ.

Proposition 3 assumes that g is strictly decreasing, which is close to assuming the conclusion. The assumption is discharged for a natural parametric family.

Proposition 4 (Closed-form interior optimum)**.** *Let $b(d) = b_0 d^\beta$ with $b_0 > 0$ and $\beta \in (0, 1]$, and let $\rho(d) = \rho_0 e^{-\gamma d}$ with $\gamma > 0$. Then g is strictly decreasing on $(0, \infty)$ for every $\lambda \geq 0$, so Proposition 3 applies whenever g changes sign on the interval. If $\lambda = 0$ the maximizer is*

$$d^\star = \beta/\gamma, \tag{5}$$

independent of b_0 and ρ_0.

Concave support value and exponential decay of repairability are therefore sufficient on their own, and the interior optimum is not an artifact of the monotonicity condition. The optimum moves outward when support value is closer to linear and inward when repair fidelity decays faster, which is the qualitative dependence the percentile sweeps probe. BRIDGE operationalizes this interior preference with a trimmed score band, while the experiments test lower and upper cutoffs, k, distance normalization, candidate-pool size, and diversity sampling.

Finally, consider a separable budget objective $U(\mathbf{m}) = \sum_j U_j(n_j + m_j)$, where each U_j is increasing and discretely concave and $\sum_j m_j = B$.

Proposition 5 (Adaptive allocation under diminishing returns)**.** *For integer allocations, repeatedly assigning one repair to the region with the largest current marginal gain maximizes $U(\mathbf{m})$. A fixed allocation can become suboptimal after a region's marginal gain falls below that of another region.*

The result follows by selecting the B largest elements from the nonincreasing marginal-gain sequences of all regions. Recomputing the acquisition signal lets the policy follow these changing values, whereas a frozen policy continues to spend according to its initial ordering.

The same model bounds how much any acquisition rule can achieve when it is applied only once.

Proposition 6 (Single-stage insensitivity)**.** *Let* $\mathbf{m}$ *and* $\mathbf{m}'$ *be any two allocations of the same budget* B *under Equation 1, computed from the same representation and applied without re-estimation. Then*

$$\left|\mathcal{B}(\mathbf{m}) - \mathcal{B}(\mathbf{m}')\right| \;\le\; B\,\gamma \max_j p_j a_j\,(n_j+\kappa)^{-\gamma-1}. \tag{6}$$

In particular, if every region carries support at least $n_{\min}$*, the two allocations differ by at most* $B\gamma\,\max_j p_j a_j\,(n_{\min}+\kappa)^{-\gamma-1}$*, which tends to zero as* $B/n_{\min}\to 0$*.*

Under one intervention, the choice of acquisition rule is therefore bounded by the ratio of budget to existing support, and no rule can separate from any other when that ratio is small. Section 6.4 measures a budget of 2,500 generated images against a source of roughly 10,000, and finds every score band from the densest quartile to the extreme tail within 1.32 points. The separation that does appear over five stages is outside the scope of this bound, because re-estimation makes the support entering the score depend on earlier allocations, so the composite map over stages is not covered by a single application of the inequality. We state the bound because it predicts the regime in which acquisition rules are indistinguishable, and we do not claim a matching lower bound for the repeated regime. Within a candidate region, farthest-first traversal gives a factor-two approximation to metric k-center coverage (Gonzalez, 1985), providing a coverage interpretation for diversity-aware selection.

The analysis yields four requirements for a practical acquisition policy. It should estimate local support, avoid sparse regions whose samples cannot be repaired faithfully, cover the chosen region without redundant anchors, and recompute its allocation after learning from each intervention. Appendix C proves Propositions 1–5. Equation 1 and the monotonic behavior assumed for b and ρ are modeling assumptions and not consequences of those proofs. Their observable predictions are tested by the selection, repair, cycle, and policy controls in Section 6 and the appendix. The next section turns the framework into a label-free algorithm.

4 BRIDGE

4.1 Cyclic acquisition and continued pretraining

BRIDGE instantiates each requirement using quantities available without labels. Mean kNN distance estimates local support. An upper-tail mode window seeks a sparse region while excluding the most extreme scores, where faithful repair is less reliable. Farthest-point sampling limits redundancy within this region, and image-to-image diffusion supplies local variants. Continued SSL changes the representation, after which BRIDGE estimates support and allocates the next budget again.

Let $D^{(0)}=\{x_i\}_{i=1}^{N_0}$ denote an unlabeled source dataset and let f_θ be an SSL encoder trained with objective $\mathcal{L}_{\text{ssl}}$. BRIDGE alternates between pretraining and dataset repair over C cycles. At cycle c, the current encoder is $f_{\theta^{(c)}}$, the current source set is $D^{(c)}=\{x_i\}_{i=1}^{N^{(c)}}$, and the corresponding embedding set is $Z^{(c)}=\{z_i^{(c)}\}_{i=1}^{N^{(c)}}$.

The loop applies three operators after each SSL training stage, which score the data, select anchors, and repair the source set. The score operator embeds all current source images and assigns an unlabeled sparsity score. The select operator converts that score distribution into a finite anchor set under a generation budget. The repair operator uses those anchors as image-to-image conditioning inputs and appends the generated variants to the next-cycle source set. The SSL encoder is the only learned component in this loop. The OOD scoring rule is nonparametric, the selection rule has no trainable parameters, and the image generator is frozen. This interface keeps BRIDGE independent of class labels and makes it compatible with any SSL objective that can provide image embeddings.

After training on $D^{(c)}$, we embed every image and compute a nonparametric OOD score from mean kNN distance,

$$\begin{aligned} z_i^{(c)} &= f_{\theta^{(c)}}(x_i),\\ d_i^{(c)} &= \frac{1}{k}\sum_{z\in\mathcal{N}_k\left(z_i^{(c)};\,Z^{(c)}\setminus\{z_i^{(c)}\}\right)} \left\|z_i^{(c)}-z\right\|_2^2, \end{aligned} \tag{7}$$

where $\mathcal{N}_k(z;Z)$ denotes the set of the k nearest neighbors of z in Z. Deep nearest-neighbor distance is also effective for OOD detection (Sun et al., 2022). Here, large $d_i^{(c)}$ indicates that an image lies in a sparse part of the learned representation space.

4.2 Sparse and recoverable anchor selection

A natural alternative is to select the largest OOD scores directly. That rule is unstable, because the farthest samples are often artifacts, degenerate crops, or semantically incoherent outliers. Uniform sampling instead spreads budget across the whole dataset, including regions that are already well covered. We target a sparse region that is still internally coherent.

Our selector works in three steps. First, it restricts attention to the upper sparse band of the empirical OOD distribution, namely the scores between the 75th and 99th percentiles, and builds a histogram over that band. Let $b^\star$ denote the modal bin and let $m^{(c)}$ denote the center of that bin. Second, given an augmentation budget B and a candidate-pool multiplier α, we form a mode-centered candidate set of size $M = \alpha B$:

$$C^{(c)} = \underset{C \subseteq U^{(c)},\, |C|=M}{\arg\min} \sum_{i \in C} \left| d_i^{(c)} - m^{(c)} \right|, \tag{8}$$

where $U^{(c)} = \{i : q_{0.75}^{(c)} \leq d_i^{(c)} \leq q_{0.99}^{(c)}\}$ and $[N^{(c)}] = \{1, \ldots, N^{(c)}\}$. Third, we choose the final anchor set $S^{(c)}$ by greedy farthest-point sampling in embedding space from that candidate pool:

$$S^{(c)} = \mathrm{FPS}\Big(C^{(c)}, B; Z^{(c)}\Big)\,. \tag{9}$$

This keeps selection inside a stable sparse regime while discouraging near-duplicate anchors. In our implementation we use $\alpha = 4$, so the final anchors are chosen from a candidate pool that is both mode-centered and sparse. Appendix Figure 4 reports how the acquisition distribution changes across repair cycles.

4.3 Local generative repair

Given the selected set $S^{(c)}$, we generate G synthetic variants per selected image with a frozen image-to-image diffusion model g_ϕ. Each selected source image is converted back to image space and used as the conditioning image. We use an empty text prompt so that the repair signal comes from the selected anchor and not from class names or hand-written prompts. In the main method we use Stable Diffusion 3 Medium in image-to-image mode with 20 denoising steps, guidance scale 5.0, strength 0.6, and output resolution equal to the training crop size. The generation ablation replaces this operator with FLUX, using six steps and guidance scale 2.5, or with an oracle that re-adds the originally removed real source images without diffusion and so bounds what any repair operator could achieve here. We first form the generated set

$$\begin{aligned} \widetilde{D}^{(c)} &= \Big\{ g_\phi(x_i, \epsilon_j) \,\Big|\, i \in S^{(c)},\, j \in [G] \Big\}, \\ D^{(c+1)} &= D^{(c)} \cup \widetilde{D}^{(c)}, \end{aligned} \tag{10}$$

where $[G] = \{1, \ldots, G\}$. The random noise input ϵ_j indexes independent translations of the same anchor, so multiple generated images can preserve coarse visual structure while changing texture, pose, color, or local composition. We do not apply an additional generated-image filter in the main pipeline.

Together, these components implement the allocation principle from Section 3. The score estimates support deficit, the mode window provides a practical recoverability constraint, FPS approximates coverage under the finite anchor budget, and cyclic re-estimation updates the allocation after every intervention. The experiments test these choices against uniform and extreme-tail selection, alternative repair operators, and matched generator settings.

5 Experimental Setup

Source datasets. We pretrain on five unlabeled source regimes. Three are long-tailed image classification datasets, namely ImageNet-100-LT (Liu et al., 2019), CIFAR-10-LT, and CIFAR-100-LT (Krizhevsky, 2009). Two are web-source unlabeled corpora, namely PASS (Asano et al., 2021) and DiffusionDB (Wang et al., 2023). For PASS and DiffusionDB we use 10k example subsets.

Downstream evaluation. We evaluate transfer on seven benchmarks, comprising CIFAR-10-LT, CIFAR-100-LT, Stanford Cars (Krause et al., 2013), FGVC Aircraft (Maji et al., 2013), Oxford Flowers (Nilsback and Zisserman, 2008), Oxford-IIIT Pets (Parkhi et al., 2012), and ImageNet-100-LT. In the main text, we report full linear-probe comparisons, together with kNN corroboration for the source-regime comparisons and the combined ablation table. The appendix provides the complete expanded linear-probe and kNN tables for every setting.

Training protocol. Unless stated otherwise, reported numbers are mean $\pm$ standard deviation over three seeds. We use ResNet-50 as the default backbone because it is the common architecture across the SSL and long-tailed SSL baselines we compare against, enabling controlled method-level comparisons without confounding improvements from newer backbone families. We also include an architecture ablation over ResNet-18, ResNet-50, ViT-S, and ViT-B. Unless noted otherwise, BRIDGE uses $k = 100$ for OOD scoring, selects 500 images per cycle, generates 5 images per selected image, and trains for 5 cycles with 100 epochs per cycle. The cycle ablation changes the number of cycles while adjusting the per-cycle budget accordingly. DINO follows the published reference recipe with a ViT-S backbone, eight local crops at 96×96, and a bottleneck projection head; it is not run on ResNet-50, because the self-distillation recipe we compare against is defined for ViT backbones. DINOv2 and MAE are likewise ViT-only and run five cycles of 80 epochs instead of 100, since DINOv2 carries multi-crop and a patch-level objective at batch 16 with 8 accumulation steps and does not finish five 100-epoch cycles inside the wall-clock limit; both objectives use 80 so that the schedule is shared. Because the downstream tasks are heterogeneous, we emphasize per-dataset tables and use the average only as a compact summary statistic. Appendix D gives additional implementation details, runtime summaries, and asset notes.

Compared methods. We compare SimCLR, TS, SDCLR, and BRIDGE. Hybrid results that combine BRIDGE with TS or SDCLR are reported in the appendix. We also pair BRIDGE with SimCLR and MoCo v3 on ViT-S to separate backbone and objective effects.

Ablation protocol. All ablations are run on ImageNet-100-LT source pretraining with the same downstream suite and three-seed evaluation. The pretraining ablation swaps SimCLR and MoCo on the shared ResNet-50 backbone, while the objective comparison that includes DINO, DINOv2 and MAE is run separately on ViT-S so that each objective keeps the backbone family it was published with. The generation ablation compares Stable Diffusion 3, FLUX, and restoration of the originally removed real data. That third arm is an oracle, not a baseline: it returns the exact images withheld when the long tail was constructed, placing real support at the locations where the method only estimates that support is missing. It therefore measures the ceiling available to any repair operator under this budget, and a generator that approaches it is performing as well as the data allow, and is not failing to beat a trivial alternative. The selection ablation compares three upper-tail mode-window ranges, a top-tail rule that takes the most OOD examples directly, and uniform sampling. The cycle ablation varies the number of repair cycles. The architecture ablation compares ResNet-18, ResNet-50, ViT-S, and ViT-B.

6 Results

6.1 Comparison across source datasets

Table 1 reports the three label-derived long-tailed source regimes with all seven downstream tasks shown directly. BRIDGE is the strongest method on the overall average in all three regimes. Under CIFAR-100-LT source pretraining BRIDGE raises the seven-task average from 23.8 for source-matched SimCLR to 34.4 and is best on every downstream dataset. On ImageNet-100-LT, BRIDGE improves the average from 39.8 to 42.5 and remains strongest overall even though TS stays competitive on the in-domain ImageNet-100-LT column. On CIFAR-10-LT, BRIDGE improves the average from 29.1 to 34.2 and is again best overall. Across the three regimes, objective-level changes can help selectively, but repairing the source set produces the strongest broad transfer.

The appendix shows that this ranking is not specific to linear probing. Under kNN transfer, BRIDGE remains strongest on average in all three label-derived regimes, improving over SimCLR from

Table 1: Linear-probe transfer for the three label-derived long-tailed source regimes. The average is computed over all seven downstream linear-probe tasks.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	73.9 ± 0.5	46.8 ± 0.4	13.7 ± 0.3	19.6 ± 0.2	35.9 ± 7.2	38.6 ± 1.5	50.4 ± 0.5	39.8 ± 1.5
	TS	72.5 ± 0.4	44.0 ± 0.5	12.5 ± 1.2	15.0 ± 0.1	35.7 ± 1.8	40.6 ± 2.2	$\mathbf{56.7 \pm 1.2}$	39.6 ± 1.1
	SDCLR	41.1 ± 1.7	18.0 ± 0.6	1.4 ± 0.3	2.3 ± 0.5	10.8 ± 1.7	10.4 ± 0.3	21.9 ± 0.6	15.1 ± 0.8
	BRIDGE (ours)	$\mathbf{75.8 \pm 0.2}$	$\mathbf{49.3 \pm 0.2}$	$\mathbf{16.4 \pm 1.2}$	$\mathbf{22.4 \pm 0.4}$	$\mathbf{36.8 \pm 1.9}$	$\mathbf{42.9 \pm 1.0}$	53.8 ± 0.1	$\mathbf{42.5 \pm 0.7}$
CIFAR-10-LT	SimCLR	67.8 ± 0.2	38.0 ± 1.0	6.5 ± 0.4	16.0 ± 0.9	$\mathbf{19.2 \pm 1.6}$	24.0 ± 1.3	32.0 ± 0.7	29.1 ± 0.9
	TS	68.5 ± 1.3	36.0 ± 1.7	4.3 ± 0.2	10.3 ± 0.1	13.1 ± 2.0	19.7 ± 2.3	31.5 ± 1.2	26.2 ± 1.3
	SDCLR	39.6 ± 1.1	15.6 ± 0.2	1.1 ± 0.1	2.4 ± 0.9	12.0 ± 1.6	9.1 ± 1.8	16.5 ± 1.6	13.8 ± 1.1
	BRIDGE (ours)	78.1 ± 0.3	$\mathbf{49.8 \pm 0.4}$	$\mathbf{9.2 \pm 0.8}$	$\mathbf{17.7 \pm 1.0}$	18.2 ± 1.1	$\mathbf{27.4 \pm 2.7}$	39.0 ± 0.6	$\mathbf{34.2 \pm 1.0}$
CIFAR-100-LT	SimCLR	59.0 ± 1.5	31.1 ± 1.6	4.1 ± 0.3	11.2 ± 1.0	12.6 ± 3.4	20.1 ± 1.0	28.7 ± 1.3	23.8 ± 1.4
	TS	56.3 ± 0.2	28.0 ± 0.1	2.8 ± 0.6	8.5 ± 0.2	15.5 ± 2.5	16.2 ± 3.1	26.2 ± 1.8	21.9 ± 1.2
	SDCLR	37.1 ± 0.8	15.7 ± 0.2	1.9 ± 0.1	2.4 ± 0.7	9.4 ± 1.0	9.9 ± 1.7	16.2 ± 0.3	13.2 ± 0.7
	BRIDGE (ours)	$\mathbf{74.2 \pm 0.5}$	$\mathbf{48.1 \pm 0.4}$	$\mathbf{8.3 \pm 1.0}$	$\mathbf{18.2 \pm 0.7}$	$\mathbf{24.4 \pm 2.3}$	27.6 ± 0.7	39.8 ± 1.1	$\mathbf{34.4 \pm 1.0}$

Table 2: Linear-probe transfer for the two web-source regimes. PASS is a natural web-image corpus, while DiffusionDB is fully synthetic. The average is computed over all seven downstream linear-probe tasks.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
PASS-10k	SimCLR	72.3 ± 0.2	45.5 ± 0.5	11.0 ± 0.2	17.5 ± 1.4	33.3 ± 1.3	33.3 ± 0.9	44.7 ± 0.9	36.8 ± 0.8
	TS	71.5 ± 0.2	43.5 ± 0.4	10.2 ± 1.2	17.2 ± 0.2	33.1 ± 2.1	36.4 ± 1.8	49.4 ± 0.4	37.3 ± 0.9
	SDCLR	42.2 ± 1.1	17.7 ± 0.8	1.8 ± 0.3	2.2 ± 0.8	10.6 ± 1.3	8.2 ± 1.6	20.8 ± 0.4	14.8 ± 0.9
	BRIDGE (ours)	$\mathbf{73.9 \pm 0.3}$	$\mathbf{47.1 \pm 0.4}$	$\mathbf{13.6 \pm 0.7}$	$\mathbf{20.5 \pm 0.7}$	$\mathbf{37.7 \pm 4.0}$	$\mathbf{37.1 \pm 0.7}$	49.1 ± 0.5	$\mathbf{39.9 \pm 1.0}$
DiffusionDB-10k	SimCLR	74.4 ± 0.5	48.6 ± 0.2	10.9 ± 0.7	17.6 ± 0.8	31.2 ± 2.4	36.1 ± 1.3	45.3 ± 0.5	37.7 ± 0.9
	TS	72.3 ± 0.3	44.4 ± 0.3	10.7 ± 0.6	15.4 ± 0.6	33.1 ± 1.1	$\mathbf{39.9 \pm 2.7}$	48.8 ± 0.2	37.8 ± 0.8
	SDCLR	42.4 ± 0.2	18.4 ± 0.2	1.1 ± 0.2	2.4 ± 0.1	13.0 ± 3.7	8.2 ± 1.4	21.0 ± 0.6	15.2 ± 0.9
	BRIDGE (ours)	$\mathbf{75.9 \pm 0.4}$	$\mathbf{49.9 \pm 0.5}$	$\mathbf{12.8 \pm 1.0}$	$\mathbf{20.0 \pm 0.4}$	$\mathbf{35.7 \pm 1.1}$	37.7 ± 0.8	49.1 ± 0.5	$\mathbf{40.1 \pm 0.7}$

31.8 to 33.6 on ImageNet-100-LT, from 24.0 to 28.7 on CIFAR-10-LT, and from 20.9 to 27.0 on CIFAR-100-LT.

Table 2 (linear-probe) evaluates the two web-source regimes, which are qualitatively different from the label-derived long-tailed datasets. PASS is a natural web image corpus, while DiffusionDB is fully synthetic. BRIDGE is again the strongest average method in both settings. On PASS-10k, it improves the seven-task average over source-matched SimCLR from 36.8 to 39.9. On DiffusionDB-10k, it improves the average from 37.7 to 40.2. Because the DiffusionDB source data are already synthetic, the gains cannot be attributed only to harvesting more natural web diversity. They instead indicate that the repair loop remains useful even when the source corpus itself carries strong generative and stylistic bias.

The same conclusion holds under the appendix's nonparametric evaluation. On PASS-10k, BRIDGE improves the seven-task kNN average from 29.3 to 30.6. On DiffusionDB-10k, it improves the average from 28.2 to 29.9. Together with Table 2, this places the effect outside both the learned probe and the natural-image setting.

6.2 Evaluation beyond the default protocol

Table 3 tests whether the result is specific to SimCLR with ResNet-50 or to frozen-feature evaluation. Five objectives are run paired on ViT-S under one schedule and one repair budget, covering contrastive learning (SimCLR), momentum contrast (MoCo v3), self-distillation (DINO, DINOv2) and masked reconstruction (MAE). Counting linear-probe and kNN accuracy on all seven downstream datasets gives 14 measurements per objective. BRIDGE improves 12 of 14 for SimCLR, 12 of 14 for MoCo v3, 13 of 14 for DINO and 12 of 14 for MAE, with mean changes of $+1.32$, $+1.31$, $+1.88$ and $+1.38$ points and a two-sided sign test at $p \leq .013$ in each case. The largest single gains are on Pets, $+7.06$ points for DINO and $+4.80$ for SimCLR under linear probing. Under kNN, MoCo v3 loses 3.88 and 3.74 points on CIFAR-10 and CIFAR-100 while gaining on the other five datasets, which is the one place where a single objective moves substantially in both directions.

DINOv2 is the exception. It improves 7 of 14 measurements, the mean change is -0.04 points, and the sign test is $p = 1.0$, so the repair loop leaves it where it started. Its baseline is also the weakest of the five, at 18.68 percent on ImageNet-100-LT against 41.95 to 44.79 for the others, because

Table 3: Evaluation beyond the default SimCLR/ResNet-50 linear-probe setting. Panel A reports linear-probe accuracy on four of the seven downstream datasets for five paired ViT-S objectives, three seeds each; Table 24 carries all seven and Table 25 the kNN protocol. Panel B reports three paired seeds for low-label full fine-tuning and PASCAL VOC semantic segmentation.

A. Frozen ViT-S transfer

Objective	Method	Cars	Aircraft	Flowers	IN100-LT
SimCLR	SSL baseline	7.15 ± 0.19	10.92 ± 0.40	32.90 ± 2.72	41.95 ± 1.43
	With BRIDGE	$\mathbf{8.18 \pm 0.42}$	$\mathbf{11.82 \pm 0.43}$	$\mathbf{36.60 \pm 2.85}$	$\mathbf{43.93 \pm 1.09}$
MoCo v3	SSL baseline	7.57 ± 0.23	11.05 ± 1.31	30.28 ± 1.36	44.79 ± 1.17
	With BRIDGE	$\mathbf{9.55 \pm 0.34}$	$\mathbf{13.42 \pm 0.98}$	$\mathbf{36.82 \pm 6.07}$	$\mathbf{47.55 \pm 0.86}$
DINO	SSL baseline	8.10 ± 0.25	11.40 ± 1.03	$\mathbf{38.13 \pm 5.24}$	44.63 ± 1.47
	With BRIDGE	$\mathbf{9.49 \pm 0.88}$	$\mathbf{13.28 \pm 0.58}$	37.25 ± 2.61	$\mathbf{46.78 \pm 0.77}$
DINOv2	SSL baseline	$\mathbf{2.13 \pm 0.35}$	$\mathbf{3.32 \pm 0.47}$	$\mathbf{14.16 \pm 2.64}$	18.68 ± 1.21
	With BRIDGE	2.06 ± 0.16	3.25 ± 0.15	12.42 ± 3.46	$\mathbf{19.35 \pm 1.34}$
MAE	SSL baseline	1.54 ± 0.19	2.44 ± 1.00	$\mathbf{10.46 \pm 5.88}$	19.93 ± 2.13
	With BRIDGE	$\mathbf{1.56 \pm 0.43}$	$\mathbf{3.18 \pm 0.70}$	8.50 ± 2.26	$\mathbf{22.19 \pm 2.35}$

B. End-to-end downstream adaptation

Evaluation	Setting	SSL baseline	With BRIDGE	Difference
Full fine-tuning	IN100-LT, 1% labels	14.31 ± 0.33	16.29 ± 1.11	$+1.97 \pm 0.78$
Full fine-tuning	IN100-LT, 10% labels	21.56 ± 0.75	24.72 ± 1.75	$+3.17 \pm 1.66$
Segmentation	PASCAL VOC, mIoU	12.78 ± 0.79	15.20 ± 0.86	$+2.42 \pm 0.27$

the published recipe carries multi-crop and a patch-level objective at batch 16 with 8 accumulation steps, and five cycles of 80 epochs at that batch size reach a point far short of where the reference recipe converges. Whether repair is neutral for this objective, or neutral for an encoder this far from convergence, is not separable from these runs. DINO carries the same self-distillation objective and improves 13 of 14, so the boundary does not fall between objective families.

The DINO arm was rebuilt for this version after we found that our earlier implementation omitted the bottleneck projection head and applied solarization to the wrong crops. Under the corrected recipe the DINO baseline is competitive with the SimCLR and MoCo v3 baselines under the same schedule, and the earlier report of a baseline at roughly half their transfer accuracy no longer stands. An earlier version of this paper reported DINO as a null on two seeds; that reading came from encoders published before the checkpointing fault described alongside Table 4 was found, and it does not survive the corrected runs.

The same source representations improve after end-to-end adaptation. With 1 and 10 percent of the ImageNet-100-LT labels, full fine-tuning gains 1.97 ± 0.78 and 3.17 ± 1.66 points. PASCAL VOC semantic segmentation improves from 12.78 ± 0.79 to 15.20 ± 0.86 mIoU, a paired gain of 2.42 ± 0.27 over three seeds.

6.3 Ablations

We now examine each design choice separately on ImageNet-100-LT source pretraining. Each ablation changes one component of BRIDGE while holding the rest of the pipeline fixed. We summarize the principal findings below and provide complete per-task linear-probe and kNN tables in the appendix. The goal is to identify which parts of the repair loop matter most, and whether the strongest configuration is driven by one isolated choice or by a coherent combination of choices.

SSL objective and backbone. Reviewers of an earlier version noted that swapping the objective on a fixed ResNet-50 confounds the objective with an architecture mismatch, since self-distillation methods are normally reported with ViT backbones. We have separated the two factors. The appendix retains the ResNet-50 comparison for SimCLR and MoCo, both of which were published with that backbone, and the objective comparison involving DINO is run on ViT-S with the published multi-crop recipe. Table 3 reports the paired full-schedule ViT-S results, in which SimCLR isolates the backbone change and MoCo v3, DINO, DINOv2 and MAE additionally change the objective. Each arm is therefore evaluated in the pairing its authors intended, and no row in this paper reports a self-distillation objective on a convolutional backbone.

Generation mechanism. Under matched sampling settings, the three-seed rerun no longer supports a ranking between FLUX.1-schnell and SD3, so we treat generator choice as an implementation choice and not as a contribution. The real-restoration condition returns images removed when constructing the long tail and serves as an oracle and not as a duplication baseline. It has access to information the method does not: the true identity and location of the missing support. Generative repair coming close to it is therefore the strong outcome, not a weak one, because it says the synthetic variants recover most of the benefit of the withheld real images while requiring no access to them. We accordingly read the small remaining gap as the cost of estimating where support is missing, not as evidence that generation is unnecessary. Separate duplicate and conventional-augmentation controls test whether reweighting anchors is sufficient.

Is the gain transfer from the generator? A frozen generator trained on external data carries information the source corpus does not, so the improvement could in principle reflect distillation from that generator and not the acquisition rule. The selection ablation separates the two accounts, because the generator is held fixed across its arms. Uniform sampling, top-tail sampling and the three mode-window variants all call the same frozen Stable Diffusion 3 checkpoint at the same sampling settings, add the same number of generated images, and receive the same number of optimizer updates. The only quantity that varies is which anchors condition the generation. Uniform acquisition reaches 35.5 ± 1.2 on the seven-task linear-probe average and direct top-tail acquisition reaches 34.3 ± 1.2, while the three mode windows reach 42.1 ± 0.7, 42.2 ± 0.9 and 42.3 ± 0.7; every mode window stays ahead of both alternatives after Holm correction, with adjusted $p \leq .011$. An account in which the external generator supplies the benefit predicts no spread between these five conditions, since all five distil from the same model at the same volume. The observed spread of 6.6 to 8.0 points is therefore attributable to the placement of the budget. This does not rule out a generator contribution to the overall level of the results, and the appendix reports direct generator-feature transfer as a separate control, but the component this paper contributes is not reducible to access to Stable Diffusion 3.

Cost of the loop. Each cycle adds one embedding pass over the current source set and one diffusion sampling stage. Measured against the corresponding no-repair run under an identical update budget, the repair loop raises wall-clock time by 29 to 46 percent across the three source regimes, with diffusion sampling accounting for most of that increase. This is the practical cost of adopting the method, and it grows with the anchor budget and the number of generated variants per anchor and not with the number of cycles at fixed total budget. Appendix E reports per-regime figures.

Feedback and alternative repair pipelines. Table 4 uses a separate matched three-stage protocol to isolate how the budget is allocated and how selected regions are repaired. Every arm now runs on encoders written after the checkpointing fault described below was fixed, and a no-repair control has been added, so the table reports what each allocation buys against spending nothing. Nothing in it separates. The seven-task linear-probe averages of the eight completed arms span 33.18 to 35.26, a range of 2.09 points against a mean seed deviation of 1.34 within a single arm. Adaptive BRIDGE reaches 34.48, the frozen selector 34.14 and no repair at all 34.28, so the schedule parameter and the repair itself are both inside the noise at three stages. Under kNN the eight arms lie between 27.46 and 27.66, a range of 0.20 points. This protocol therefore measures the allocation question in the regime where Section 6.4 predicts no rule can separate from any other, and its function here is to bound that regime and not to rank the pipelines. The five-cycle experiments in Table 3 and the selection ablation are where repair is shown to pay. The adaptive-versus-frozen contrast in this table required a rerun. In our earlier runs the frozen-selector arm reproduced the adaptive encoder to the last bit, which we initially read as the schedule parameter being inert. It was a checkpointing fault: each cycle constructs a fresh trainer whose step counter restarts at zero and stops at the same step limit, and the checkpoint callback, which is reused across cycles, treats a save at a step it has already written as a duplicate and skips it. Every run therefore published its first-cycle encoder no matter how the later cycles diverged. Anchor selection had in fact diverged, with the adaptive and frozen manifests overlapping on 14 of 500 anchors from the second cycle onward. The rows reported here come from reruns with the callback state reset between cycles, and we verify before publication that no two conditions in this table share an encoder fingerprint.

Sample selection. Table 5 compares three upper-tail mode-window variants with top-tail and uniform sampling. All three diversity-aware mode-window variants are much stronger. The q50–99 and q1–99 variants remain close to the default, showing that the exact lower cutoff is not critical once selection is restricted to a sparse band and diversified within it. We use q75–99 as a fixed default and do not claim it is universally optimal.

Reading the robustness sweep. Appendix Table 21 varies the candidate-pool multiplier, the upper cutoff, the neighbourhood size, the distance and the selection strategy one at a time. The table contains a built-in scale for its own resolution. Cosine scoring rescales the normalized L_2 score by a positive constant, which leaves every ranking and every quantile position unchanged, so the cosine

Table 4: Matched feedback and repair controls on ImageNet-100-LT. All methods start from paired source checkpoints and receive three 60-epoch stages with the same update cap. Generation methods add 7,500 images in total. A no-repair arm receives the same optimizer budget and adds nothing. Each entry is downstream linear-probe accuracy as mean ± standard deviation over three seeds; Table 26 carries all seven datasets and Table 27 the kNN protocol.

Method	Cars	Aircraft	Flowers	IN100-LT
No repair	12.33 ± 0.53	16.49 ± 1.01	28.76 ± 1.96	42.56 ± 0.74
Adaptive BRIDGE	12.68 ± 0.88	17.14 ± 0.92	27.89 ± 4.99	42.34 ± 0.69
Frozen first-cycle selector	12.70 ± 0.78	16.89 ± 0.98	25.93 ± 4.73	43.64 ± 0.45
One-shot repair	12.73 ± 0.43	$\mathbf{18.13 \pm 0.46}$	$\mathbf{28.98 \pm 2.72}$	$\mathbf{43.66 \pm 0.76}$
Uniform acquisition	$\mathbf{12.83 \pm 0.42}$	17.73 ± 0.80	27.23 ± 6.07	43.10 ± 0.78
Top-tail acquisition	12.12 ± 0.35	17.18 ± 0.90	27.89 ± 2.64	43.27 ± 0.77
Mode-window conventional augmentation	12.04 ± 0.18	17.01 ± 1.05	22.66 ± 4.35	41.28 ± 0.46
AIDE-style VLM acquisition and text-to-image	12.28 ± 0.15	17.86 ± 0.55	28.76 ± 0.65	43.36 ± 0.08
Mode-window caption and text-to-image	12.42 ± 1.13	17.51 ± 1.47	25.05 ± 3.72	43.64 ± 0.79
Mode-window captioned image-to-image	12.66 ± 0.56	17.59 ± 0.30	25.27 ± 2.95	43.13 ± 0.61

and normalized L_2 rows are the same experiment executed twice. They differ by up to 1.3 points on individual targets. Differences of that size elsewhere in the table are therefore not readable as effects of the knob being varied, and the useful conclusion from the sweep is the absence of any variation exceeding this floor, and not an ordering within it. The contrast with the 6.6 to 8.0 point gap to uniform and top-tail acquisition is what identifies the sparse band as the component that carries the result.

Cycle count. Performance improves sharply over several repair rounds. Relative to 2 cycles, 5 cycles improve the seven-task averages by 7.3 points for linear probing and 5.7 points for kNN, while remaining stronger than 10 cycles. Longer schedules therefore do not help monotonically in this regime.

Table 5: Selection and cycle ablations on ImageNet-100-LT. This matched ablation family is independent of the source-regime runs in Table 1. Values average all seven downstream tasks over three seeds. The nearby mode-window results show that the exact lower percentile is not critical. Uniform and top-tail selection test the dense-allocation and extreme-sparsity alternatives.

Group	Setting	Linear probe	kNN
Selection	Mode-window q1–99	42.1 ± 0.7	33.4 ± 0.3
	Mode-window q50–99	42.2 ± 0.9	33.6 ± 0.3
	Mode-window q75–99	42.3 ± 0.7	33.6 ± 0.5
	Uniform	35.5 ± 1.2	28.4 ± 0.6
	Top-tail	34.3 ± 1.2	27.3 ± 0.6
Schedule	2 cycles	35.0 ± 1.4	27.9 ± 0.5
	5 cycles	42.3 ± 0.7	33.6 ± 0.5
	10 cycles	34.1 ± 1.7	28.1 ± 0.6

Encoder architecture. ResNet-50 gives the strongest seven-task linear-probe average at 42.3 and leads on six targets. ViT-B is best on Flowers but averages 34.8, making ResNet-50 the preferred backbone in this sweep.

kNN corroboration. The appendix reports nonparametric transfer for every ablation. Mode-window selection remains stronger than top-tail and uniform selection over five cycles, and five cycles remains stronger than shorter schedules. Paired ViT and matched FLUX experiments provide the controlled objective and generator comparisons.

Table 6: Repair utility by sparsity band on ImageNet-100-LT. Each row spends the identical acquisition budget on a different quantile range of the kNN score distribution, holding every other part of the loop fixed. Mean $\pm$ standard deviation over three seeds. Linear-probe accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
q0–25 (densest)	6.73 ± 0.36	9.63 ± 0.26	24.84 ± 5.10	23.37 ± 2.72	62.97 ± 0.82	33.96 ± 0.49	34.44 ± 0.62
q25–50	6.80 ± 0.39	10.28 ± 1.37	23.75 ± 2.10	22.74 ± 1.57	62.76 ± 0.64	33.20 ± 1.12	35.61 ± 0.23
q50–75	6.79 ± 0.63	10.24 ± 1.94	20.04 ± 4.64	23.55 ± 2.58	62.24 ± 0.37	32.80 ± 1.66	34.51 ± 1.27
q75–85	6.12 ± 0.33	9.74 ± 1.00	22.88 ± 4.58	23.46 ± 1.64	60.97 ± 1.05	31.90 ± 0.36	33.55 ± 0.46
q85–90	6.62 ± 0.39	9.95 ± 0.73	23.75 ± 3.09	25.45 ± 1.77	63.38 ± 0.78	33.47 ± 0.93	35.24 ± 0.69
q90–95	6.55 ± 0.55	10.46 ± 1.87	22.66 ± 3.94	25.27 ± 0.72	62.30 ± 1.15	33.49 ± 0.87	34.40 ± 0.62
q95–97	6.58 ± 0.20	9.55 ± 1.62	23.09 ± 1.00	23.55 ± 0.68	61.77 ± 2.13	32.75 ± 2.27	34.99 ± 1.19
q97–99	6.80 ± 0.17	11.38 ± 2.82	20.26 ± 3.98	26.27 ± 3.24	62.73 ± 1.49	33.13 ± 1.97	35.31 ± 1.55
q99–100 (extreme tail)	6.55 ± 0.28	10.78 ± 1.23	21.79 ± 0.75	23.82 ± 1.37	63.54 ± 0.92	33.65 ± 0.55	35.57 ± 0.91

6.4 WHERE THE ACQUISITION RULE ACQUIRES ITS EFFECT

Table 6 spends one identical acquisition budget on each decile-scale band of the kNN score distribution, from the densest quartile to the extreme tail, and changes nothing else in the loop. The seven-task linear-probe averages of the nine bands lie between 26.95 and 28.27. The spread over the whole score range is thus 1.32 points, which is the same size as the 1.3-point separation between the two rows of the robustness table that are provably the identical experiment. One repair stage does not distinguish the densest quartile of the representation from its extreme tail. The selector sweep in Appendix Table 21 behaves the same way, and it also uses one repair stage.

Under five repair stages the same contrast is large. Mode-window acquisition reaches 42.3 ± 0.7, uniform acquisition 35.5 ± 1.2 and top-tail acquisition 34.3 ± 1.2, a spread of 8.0 points across rules that are indistinguishable after a single intervention. The two protocols differ in the number of times the acquisition signal is recomputed, and not in the acquisition rules, the total repair volume or the total optimizer budget.

We read this as the central empirical result of the paper. The value of a targeted acquisition rule is not delivered by the individual intervention it makes; a single well-placed batch of 2,500 generated images is worth no more than a badly placed one. The value accumulates over the loop, because each intervention changes the representation that chooses the next one, and small differences in where support is added compound into large differences in what the encoder can score. This is the qualitative behavior that Proposition 5 describes, and it separates BRIDGE from a one-shot selection rule with the same components. It also sets a limit on the theory: Proposition 3 predicts an interior optimum in the acquisition score, and at one repair stage that prediction is not detectable in our setting. The interior preference is visible in the repeated regime and not in the single-shot regime, so we present the propositions as a description of the policy class and not as a claim about any individual repair.

6.5 REPRESENTATION ANALYSIS

We use a fixed panel of original images to keep the evaluation population unchanged across checkpoints. With MoCo v3 and ViT-S, effective rank increases by 21.97 ± 1.58, spectral entropy by 0.296 ± 0.019, and 1-NN accuracy by 3.80 ± 0.80. The rank and entropy changes remain significant after Holm correction. SimCLR improves transfer while its global effective rank decreases, so increasing global rank is not a universal account of the method.

We separately test the local intervention in the frozen cycle-0 representation. First-cycle anchors are matched one-to-one within class to untreated images with the same initial kNN radius. After inserting generated samples, the radius falls more around treated anchors in every seed, with a mean difference-in-differences of -0.00134 ± 0.00067. Neighborhood purity changes by $+0.0134 \pm 0.0032$ relative to the matched controls. This verifies that the repair operator places support near the selected regions. It does not by itself establish that subsequent SSL must preserve that local geometry, and the learned-space anchor comparison is mixed for SimCLR. We therefore use the transfer results as the primary measure of utility and report the full fixed-space and learned-space analyses in the appendix.

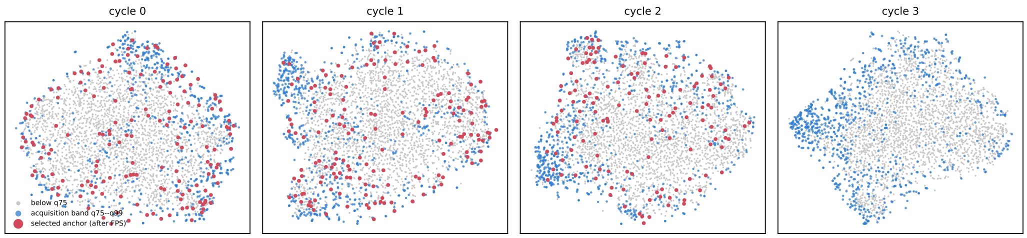

Figure 2: Where the acquisition budget is spent, projected to two dimensions with t-SNE, for one ImageNet-100-LT run. Grey marks scores below the 75th percentile, blue the q75–q99 acquisition band, and red the anchors farthest-point sampling selects from that band. The band occupies the periphery of the projection: over the panel shown, the rank correlation between kNN score and distance from the projection centre is $+0.51$, and band members sit at mean radius 34.6 against 25.5 for the rest. The anchors are spread across the band and not clustered inside it, which is the behaviour farthest-point sampling is included to produce. Each panel is projected separately because the encoder changes between cycles, so positions are comparable within a panel and not across panels. The panel contains original images only, so it shows which regions are targeted and not what the inserted samples do to them; Table 8 measures the latter directly.

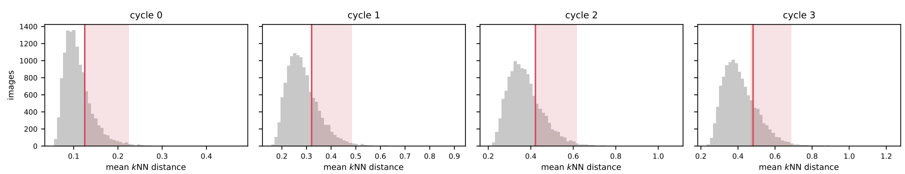

Figure 3: Score distribution over repair cycles for the same run, with the q75–q99 acquisition band shaded and the modal bin marked. The distribution stays right-skewed while its scale grows, so the band tracks a moving target and the selector re-derives it from the current representation at every cycle.

7 Scope and Limitations

BRIDGE depends on informative encoder neighborhoods and faithful local generation. Errors in either component can feed artifacts into later stages, while the frozen generator imports biases from its pretraining data. Its score band, neighborhood size, and budget are operating choices and not universal constants. SimCLR, MoCo v3, DINO and MAE all improve on 12 or 13 of 14 measurements, while DINOv2 moves by -0.04 points and does not, so the benefit is not uniform across SSL objectives. DINO shares the self-distillation objective with DINOv2 and does improve, which leaves the boundary unexplained by the objective family. The DINOv2 baseline is also the weakest of the five by a factor of two, and our schedule stops well short of where its published recipe converges, so these runs cannot separate an objective that resists repair from an encoder too far from convergence for repair to act on. A study that trains DINOv2 to convergence before repairing is left open. The fixed-space analysis verifies where support is inserted, not how every learned neighborhood must evolve. Finally, a cycle adds an embedding pass and generation, raising measured wall-clock time by 29–46 percent. Corpus-scale effectiveness beyond our controlled 10k web subsets remains open and may require a growing repair budget.

8 Conclusion

We formulate skewed-data SSL as a feedback problem in which the learner allocates a finite budget to change its future source distribution. Recoverability-weighted support utility links sparse-region repair, interior acquisition, diversity, and changing marginal value after intervention. BRIDGE realizes these principles through a label-free loop that improves transfer more reliably than uniform addition, maximal-outlier generation, one-shot generation, and the tested language-mediated alternatives. The comparison between one repair stage and five locates the source of that improvement. Where the budget is spent has no measurable consequence for a single intervention, and it has an 8.0-point consequence once the representation that chooses the target is itself repeatedly updated by the repairs. A selection heuristic evaluated outside such a loop would therefore have appeared to be worthless. Repeated source repair offers a practical complement to improving the SSL objective alone.

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A Comparison with Related Acquisition Methods

Table 7 makes the comparison axes explicit. Classical oversampling and TADA receive labels or supervised learning signals. DOPING uses latent rarity but targets anomaly detection and does not close the loop through a changing SSL representation. GenRep and StableRep make synthetic images or latent neighborhoods the source of contrastive views without allocating a repair budget over an observed skewed corpus. DiffAug is the closest iterative SSL antecedent, but it jointly learns a generator for generic positive views and does not ask where a frozen generator should intervene.

AIDE and industrial data-engine systems establish the broader model–data feedback pattern, while using supervised task failures, VLM/LLM reasoning, annotation, and domain-specific verification. BRIDGE studies the label-free special case in which local support in the current SSL representation is both the diagnostic and acquisition signal. Our VLM rare-concept baseline adapts the AIDE idea by clustering frozen CLIP features, selecting inversely to cluster occupancy, captioning the selected images, and generating from text. The paired text-to-image and captioned SDEdit controls separately test whether retaining the selected image as a local condition matters.

Table 7: Closest methods and the dimensions relevant to the acquisition problem. "Feedback" means that the learner or task model changes the next acquisition decision.

Method	Labels or task loss	Acquisition signal	Data action	Feedback	Primary goal
SMOTE (Chawla et al., 2002)	class labels	class minority	interpolate examples	no	supervised imbalance
DOPING (Lim et al., 2018)	no labels	latent edge density	latent oversampling	no	anomaly detection
Concept pruning (Abbas et al., 2024)	no task labels	cluster complexity	remove examples	no	efficient pretraining
DA-Fusion (Trabucco et al., 2024)	labels / few-shot set	class and exemplar	image-conditioned diffusion	no	supervised recognition
TADA (Nguyen et al., 2026)	supervised loss dynamics	slow-learned examples	faithful diffusion variants	no	supervised recognition
GenRep / StableRep (Jahanian et al., 2022; Tian et al., 2023)	prompts or generator latent	global sampling	synthetic SSL views	no	representation learning
DiffAug (Zang et al., 2024)	no labels	learned positive-view objective	jointly learned diffusion views	yes	contrastive learning
AIDE (Liang et al., 2024)	task labels and failures	VLM/LLM failure analysis	curate, generate, annotate	yes	driving detection
BRIDGE	no labels	local support in current SSL space	budgeted local variants	yes	source-support repair

The generator literature also identifies the origins of the repair operator and its controls. SDEdit supplies the image-editing mechanism, while synthetic-recognition studies establish the importance of prompts, filtering, diversity, and domain gap. DreamTeacher and DIFT motivate direct generator-feature transfer, and REPA studies the reverse direction from a visual encoder into a diffusion model. Omniverse Replicator provides non-archival systems context for iterative synthetic-data pipelines. BRIDGE brings these tools into a label-free local acquisition problem with a finite intervention budget and repeated re-estimation.

B Representation Diagnostics

The fixed-space intervention test freezes the cycle-0 encoder and queries the same original images before and after inserting all generated samples. First-cycle anchors are matched one-to-one within class to original images with the closest initial kNN radius that are never selected in any cycle. Table 8 reports the direct anchor-versus-control comparison. The learned-space comparison retrains the encoder after the intervention and evaluates the same original-image panel. Its local effect is mixed for SimCLR, while the global response depends on the objective. We therefore treat the fixed-space result as evidence about where new support is inserted, not as evidence that all learned representations must change in the same geometric direction.

C Proofs for the Support-Allocation Framework

Proof of Proposition 1. Sort the distances from z to points in Z and denote their kth order statistic by $r_k(z; Z)$. Adding A preserves every old candidate distance and may introduce smaller ones. The kth order statistic can therefore only decrease. It decreases strictly when an added distance enters the first k positions and displaces a strictly larger distance.

Proof of Proposition 2. Let $x = n_j + m_j + \kappa$. Since $f(x) = x^{-\gamma}$ is positive, strictly decreasing, and strictly convex for $x > 0$, the forward difference $f(x) - f(x+1)$ is positive and strictly decreases with x. Multiplication by $p_j a_j > 0$ preserves both properties. Equal $p_j a_j$ therefore leaves effective support as the sole determinant of the marginal ordering.

Proof of Proposition 3. Rewrite the utility as $u(d) = \rho(d)(b(d) + \lambda) - \lambda$. Its derivative is

$$u'(d) = \rho(d)(b(d) + \lambda) \left[\frac{b'(d)}{b(d) + \lambda} + \frac{\rho'(d)}{\rho(d)} \right] = \rho(d)(b(d) + \lambda) g(d).$$

The factor preceding $g(d)$ is positive. Since g is strictly decreasing and changes sign across the interval, it has one root $d^\star$ by continuity. Hence u increases before $d^\star$ and decreases after it, making $d^\star$ the unique global maximizer.

Table 8: Local support added in the frozen cycle-0 representation, reported as mean $\pm$ standard deviation over three seeds. Controls are matched one-to-one within class on initial kNN radius and are never selected. Negative radius DiD and positive purity DiD favor treated anchors.

	Δr anchor	Δr control	Radius DiD	Purity DiD
Mean $\pm$ SD	-0.00273 ± 0.00062	-0.00139 ± 0.00004	-0.00134 ± 0.00067	$+0.0134 \pm 0.0032$

Proof of Proposition 4. With $b(d) = b_0 d^\beta$ and $\rho(d) = \rho_0 e^{-\gamma d}$ we have $\rho'(d)/\rho(d) = -\gamma$, so

$$g(d) = \frac{b_0 \beta d^{\beta-1}}{b_0 d^\beta + \lambda} - \gamma = \frac{\beta d^{\beta-1}}{d^\beta + c} - \gamma, \qquad c = \lambda / b_0 \geq 0.$$

Write $h(d) = \beta d^{\beta-1}(d^\beta + c)^{-1}$. Differentiating,

$$h'(d) = \frac{\beta d^{\beta-2}\big[(\beta-1)(d^\beta + c) - \beta d^\beta\big]}{(d^\beta + c)^2} = \frac{\beta d^{\beta-2}\big[(\beta-1)c - d^\beta\big]}{(d^\beta + c)^2}.$$

For $\beta \in (0, 1]$ we have $(\beta - 1)c \leq 0 < d^\beta$ for all $d > 0$, so the bracket is strictly negative and $h'(d) < 0$. Hence $g = h - \gamma$ is strictly decreasing for every $\lambda \geq 0$. When $\lambda = 0$ we have $c = 0$ and $g(d) = \beta/d - \gamma$, whose unique root is $d^\star = \beta/\gamma$; g is positive below it and negative above it, so $d^\star$ is the maximizer by the argument of Proposition 3.

Proof of Proposition 6. Write $f(x) = x^{-\gamma}$. For $x > 0$ the mean value theorem gives $f(x) - f(x + 1) = \gamma \xi^{-\gamma-1}$ for some $\xi \in (x, x + 1)$, so

$$0 < \Delta_j(m_j) = p_j a_j \big[f(n_j + m_j + \kappa) - f(n_j + m_j + 1 + \kappa)\big] \leq p_j a_j \gamma (n_j + \kappa)^{-\gamma-1},$$

using $\xi > n_j + m_j + \kappa \geq n_j + \kappa$ and $m_j \geq 0$. Both $\mathcal{B}(\mathbf{m})$ and $\mathcal{B}(\mathbf{m}')$ are obtained from the common baseline $\mathcal{B}(\mathbf{0})$ by subtracting sums of at most B such marginals, since $\sum_j m_j = \sum_j m'_j = B$. Each sum therefore lies in $[0,\ B\gamma \max_j p_j a_j (n_j + \kappa)^{-\gamma-1}]$, and the difference of two numbers drawn from an interval is bounded by its length. The stated bound follows, and monotonicity of $(n+\kappa)^{-\gamma-1}$ in n gives the $n_{\min}$ form.

Proof of Proposition 5. For region j, define the marginal sequence $\delta_{j,t} = U_j(n_j+t) - U_j(n_j+t-1)$ for $t \geq 1$. Discrete concavity makes each sequence nonincreasing. Any feasible allocation of B repairs selects a prefix of length m_j from each sequence. Greedy allocation repeatedly chooses the largest available next element. After B steps it has selected the B largest prefix-feasible marginals. If another allocation had larger utility, it would contain a selected marginal smaller than an available unselected marginal, and exchanging them would improve that allocation. This contradicts optimality. A frozen allocation can fail this rule because it preserves the initial ordering after selected regions move to smaller marginals.

D Implementation and Reproducibility

Core training configuration. The main experiments use ResNet-50 encoders. Most baseline, BRIDGE, and source-regime comparisons use 5 cycles with 100 epochs per cycle. The default BRIDGE configuration uses mean kNN distance with $k = 100$, selects 500 samples per cycle, and generates 5 images per selected sample. The default optimizer is AdamW with weight decay 10^{-4}, and method-specific learning rates and temperatures follow the settings used in each experiment.

OOD scoring and selection. The submitted experiments used raw squared ℓ_2 distance for OOD scoring and normalized features for farthest-point sampling. Because raw feature norms can confound cross-objective comparisons, the new controlled experiments use normalized ℓ_2 throughout and report raw ℓ_2, normalized ℓ_2, cosine, and multiscale-k variants as a robustness study. The mode-window selector restricts the OOD distribution to the 75th to 99th percentile range, builds a histogram with the code-level auto-bin heuristic, forms a mode-centered candidate pool of size $4B$, and then applies greedy farthest-point sampling to select the final B anchors.

Generators. The main method uses Stable Diffusion 3 in image-to-image mode. The generation ablation replaces it with FLUX or with a no-generation re-population baseline. In the PASS and DiffusionDB experiments, we increase the SD3 batch size to 12 to fit the 24-hour wall-time limit.

Source data specifics. PASS and DiffusionDB use `train[:10000]` with pretraining split override `[1.0, 0.0, 0.0]`. This keeps the source data budget aligned across methods while avoiding unnecessary pretraining validation overhead. The long-tailed ImageNet-100, CIFAR-10, and CIFAR-100 source regimes use their standard long-tailed source configurations, without the 10k subset restriction used for PASS and DiffusionDB.

Seeds and reporting. All main claims are supported by three seed averages. Tables report mean $\pm$ standard deviation over those seeds. The exact table values are computed directly from the three runs corresponding to each reported setting.

Code availability. An anonymous code release accompanying this submission is available at `https://anonymous.4open.science/status/FOMO-0488`.

E Compute Resources and Runtime

Most reported experiments use one node with two NVIDIA A100 80GB GPUs and a 24-hour wall-time limit. The DINO arm uses the reference multi-crop setting of two global crops and eight local crops at 96$\times$96, which costs about 3.5 global-crop forward passes per image and fits on a single 48GB card at the batch size used here. Across the successful experiments we report, the aggregate training time is approximately 3022 wall-clock hours, or about 6142 GPU hours. This excludes failed runs, timeout runs, cache migration jobs, and other debugging or pilot jobs, so the full project required more compute than the successful totals alone.

Representative mean wall-clock times over three seeds are as follows. On ImageNet-100-LT source pretraining, SimCLR takes 16.9 hours and BRIDGE takes 21.8 hours. On PASS, SimCLR takes 10.2 hours and BRIDGE takes 14.9 hours. On DiffusionDB, SimCLR takes 11.6 hours and BRIDGE takes 16.4 hours. The repair loop therefore raises wall-clock time by 29 to 46 percent over the corresponding no-repair run, of which the embedding pass accounts for a small fraction and diffusion sampling for the remainder.

F Assets and Terms of Use

We use existing datasets and pretrained generators under their original release terms.

Source datasets. ImageNet-100-LT is derived from ImageNet, whose official access terms restrict the images to non commercial research and educational use. PASS is released under Creative Commons Attribution 4.0. DiffusionDB is released under CC0-1.0. CIFAR-10, CIFAR-100, Stanford Cars, FGVC Aircraft, Oxford Flowers, and Oxford-IIIT Pets are used from their official academic benchmark releases. When an official page provides download terms instead of a standard software style license, we follow those original terms.

Generators. Stable Diffusion 3 Medium is used under the Stability AI Community License. FLUX.1-dev is used only in the ablation and is subject to the FLUX.1-dev Non-Commercial License. We do not redistribute model weights, third party datasets, or generated image corpora as part of this submission.

G Broader Impact

A method that improves representation quality under biased source data can reduce the need for manual relabeling and can make SSL more useful in domains where balanced data are difficult to collect. That is a positive outcome. At the same time, our method relies on large image generators and on web-sourced datasets. These carry familiar risks related to license compliance, privacy, and

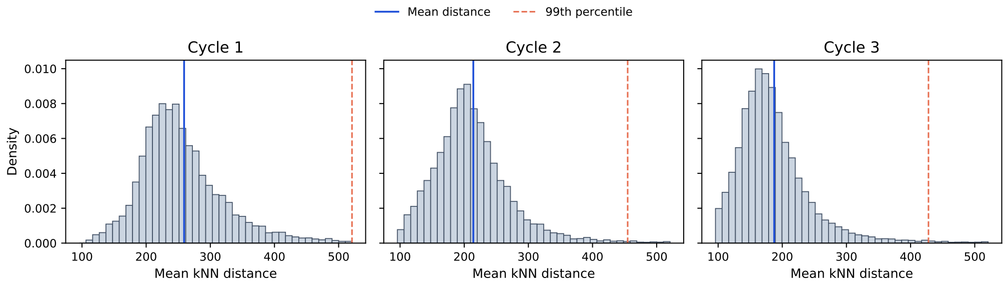

Figure 4: Representative OOD histograms across BRIDGE cycles from an ImageNet-100-LT run. Solid blue lines mark the mean distance and dashed orange lines mark the 99th percentile. The acquisition distribution and its modal shoulder change as training and repair progress.

unintended memorization or harmful content. Our experiments stay within existing dataset and model release terms and do not release new generators or scraped image collections. Any future deployment would still require a task-specific review of fairness, privacy, and content safety.

H FULL RESULTS

Table 9: Full linear-probe transfer results across all source datasets and comparison methods.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	73.9 ± 0.5	46.8 ± 0.4	13.7 ± 0.3	19.6 ± 0.2	35.9 ± 7.2	38.6 ± 1.5	50.4 ± 0.5	39.8 ± 1.5
	TS	72.5 ± 0.4	44.0 ± 0.5	12.5 ± 1.2	15.0 ± 0.1	35.7 ± 1.8	40.6 ± 2.2	**56.7 ± 1.2**	39.6 ± 1.1
	SDCLR	41.1 ± 1.7	18.0 ± 0.6	1.4 ± 0.3	2.3 ± 0.5	10.8 ± 1.7	10.4 ± 0.3	21.9 ± 0.6	15.1 ± 0.8
	BRIDGE (ours)	**75.8 ± 0.2**	**49.3 ± 0.2**	**16.4 ± 1.2**	**22.4 ± 0.4**	**36.8 ± 1.9**	**42.9 ± 1.0**	53.8 ± 0.1	**42.5 ± 0.7**
	BRIDGE+TS (ours)	74.2 ± 0.5	45.6 ± 0.8	9.4 ± 0.5	15.0 ± 0.2	25.5 ± 2.5	33.1 ± 2.8	48.3 ± 0.4	35.9 ± 1.1
	BRIDGE+SDCLR (ours)	43.8 ± 1.7	18.4 ± 0.7	2.0 ± 0.3	2.5 ± 0.8	10.0 ± 0.8	12.2 ± 3.1	21.8 ± 0.5	15.8 ± 1.1
CIFAR-10-LT	SimCLR	67.8 ± 0.2	38.0 ± 1.0	6.5 ± 0.4	16.0 ± 0.9	**19.2 ± 1.6**	24.0 ± 1.3	32.0 ± 0.7	29.1 ± 0.9
	TS	68.5 ± 1.3	36.0 ± 1.7	4.3 ± 0.2	10.3 ± 0.1	13.1 ± 2.0	19.7 ± 2.3	31.5 ± 1.2	26.2 ± 1.3
	SDCLR	39.6 ± 1.1	15.6 ± 0.2	1.1 ± 0.1	2.4 ± 0.9	12.0 ± 1.6	9.1 ± 1.8	16.5 ± 1.6	13.8 ± 1.1
	BRIDGE (ours)	78.1 ± 0.3	**49.8 ± 0.4**	**9.2 ± 0.8**	**17.7 ± 1.0**	18.2 ± 1.1	**27.4 ± 2.7**	39.0 ± 0.6	**34.2 ± 1.0**
	BRIDGE+TS (ours)	**78.6 ± 0.3**	46.1 ± 0.3	7.5 ± 0.9	11.1 ± 0.7	14.5 ± 0.6	25.1 ± 2.7	**39.6 ± 0.8**	31.8 ± 0.9
	BRIDGE+SDCLR (ours)	40.9 ± 0.7	17.9 ± 0.5	1.4 ± 0.1	2.2 ± 0.6	11.0 ± 2.4	10.0 ± 2.7	21.0 ± 0.6	14.9 ± 1.1
CIFAR-100-LT	SimCLR	59.0 ± 1.5	31.1 ± 1.6	4.1 ± 0.3	11.2 ± 1.0	12.6 ± 3.4	20.1 ± 1.0	28.7 ± 1.3	23.8 ± 1.4
	TS	56.3 ± 0.2	28.0 ± 0.1	2.8 ± 0.6	8.5 ± 0.2	15.5 ± 2.5	16.2 ± 3.1	26.2 ± 1.8	21.9 ± 1.2
	SDCLR	37.1 ± 0.8	15.7 ± 0.2	1.9 ± 0.1	2.4 ± 0.7	9.4 ± 1.0	9.9 ± 1.7	16.2 ± 0.3	13.2 ± 0.7
	BRIDGE (ours)	**74.2 ± 0.5**	**48.1 ± 0.4**	**8.3 ± 1.0**	**18.2 ± 0.7**	**24.4 ± 2.3**	27.6 ± 0.7	39.8 ± 1.1	**34.4 ± 1.0**
	BRIDGE+TS (ours)	73.7 ± 0.1	47.3 ± 0.5	7.9 ± 0.6	14.1 ± 0.5	18.2 ± 3.2	**28.4 ± 0.4**	**41.3 ± 0.3**	33.0 ± 0.8
	BRIDGE+SDCLR (ours)	42.0 ± 1.5	16.9 ± 0.9	1.3 ± 0.1	2.2 ± 0.6	10.6 ± 0.8	8.3 ± 0.8	20.2 ± 0.9	14.5 ± 0.8
PASS-10k	SimCLR	72.3 ± 0.2	45.5 ± 0.5	11.0 ± 0.2	17.5 ± 1.4	33.3 ± 1.3	33.3 ± 0.9	44.7 ± 0.9	36.8 ± 0.8
	TS	71.5 ± 0.2	43.5 ± 0.4	10.2 ± 1.2	17.2 ± 0.2	33.1 ± 2.1	36.4 ± 1.8	49.4 ± 0.4	37.3 ± 0.9
	SDCLR	42.2 ± 1.1	17.7 ± 0.8	1.8 ± 0.3	2.2 ± 0.8	10.6 ± 1.3	8.2 ± 1.6	20.8 ± 0.4	14.8 ± 0.9
	BRIDGE (ours)	**73.9 ± 0.3**	**47.1 ± 0.4**	**13.6 ± 0.7**	**20.5 ± 0.7**	**37.7 ± 4.0**	**37.1 ± 0.7**	49.1 ± 0.5	**39.9 ± 1.0**
	BRIDGE+TS (ours)	68.9 ± 0.6	40.6 ± 0.7	12.1 ± 0.4	14.1 ± 0.5	34.6 ± 2.9	31.6 ± 1.5	**52.4 ± 0.6**	36.4 ± 1.0
	BRIDGE+SDCLR (ours)	44.0 ± 0.9	18.8 ± 0.9	1.4 ± 0.3	1.8 ± 0.9	12.6 ± 2.6	12.4 ± 1.3	21.2 ± 1.1	16.0 ± 1.1
DiffusionDB-10k	SimCLR	74.4 ± 0.5	48.6 ± 0.2	10.9 ± 0.7	17.6 ± 0.8	31.2 ± 2.4	36.1 ± 1.3	45.3 ± 0.5	37.7 ± 0.9
	TS	72.3 ± 0.3	44.4 ± 0.3	10.7 ± 0.6	15.4 ± 0.6	33.1 ± 1.1	**39.9 ± 2.7**	48.8 ± 0.2	37.8 ± 0.8
	SDCLR	42.4 ± 0.2	18.4 ± 0.2	1.1 ± 0.2	2.4 ± 0.1	13.0 ± 3.7	8.2 ± 1.4	21.0 ± 0.6	15.2 ± 0.9
	BRIDGE (ours)	**75.9 ± 0.4**	**49.9 ± 0.5**	**12.8 ± 1.0**	**20.0 ± 0.4**	**35.7 ± 1.1**	37.7 ± 0.8	49.1 ± 0.5	**40.1 ± 0.7**
	BRIDGE+TS (ours)	67.8 ± 0.6	39.1 ± 0.5	9.4 ± 1.2	12.4 ± 0.4	30.3 ± 3.0	33.7 ± 1.9	**50.8 ± 0.6**	34.8 ± 1.2
	BRIDGE+SDCLR (ours)	43.8 ± 0.9	18.5 ± 0.3	1.3 ± 0.4	2.5 ± 0.0	14.3 ± 1.6	11.4 ± 0.6	21.9 ± 0.6	16.2 ± 0.6

Table 10: Full kNN transfer results across all source datasets and comparison methods.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	68.6 ± 0.4	37.3 ± 0.2	6.0 ± 0.7	11.2 ± 0.4	32.2 ± 0.7	23.6 ± 0.3	43.9 ± 0.9	31.8 ± 0.5
	TS	68.4 ± 0.3	36.5 ± 0.1	4.7 ± 0.2	8.3 ± 0.5	26.6 ± 1.0	23.6 ± 1.7	**52.2 ± 0.5**	31.5 ± 0.6
	SDCLR	32.5 ± 0.4	10.9 ± 0.2	1.0 ± 0.3	2.1 ± 0.1	10.7 ± 0.7	4.7 ± 0.2	11.2 ± 0.2	10.4 ± 0.3
	BRIDGE (ours)	**69.0 ± 0.5**	**37.8 ± 0.3**	**7.0 ± 0.2**	**11.7 ± 0.2**	**36.5 ± 0.4**	**25.3 ± 1.0**	48.1 ± 0.6	**33.6 ± 0.4**
	BRIDGE+TS (ours)	65.4 ± 0.2	33.6 ± 0.3	3.1 ± 0.1	7.1 ± 0.3	18.2 ± 0.2	15.6 ± 0.7	40.8 ± 0.5	26.3 ± 0.3
	BRIDGE+SDCLR (ours)	36.3 ± 0.8	13.0 ± 0.5	1.1 ± 0.3	2.4 ± 0.2	9.7 ± 0.3	5.4 ± 0.4	13.9 ± 1.0	11.7 ± 0.5
CIFAR-10-LT	SimCLR	66.9 ± 0.4	33.4 ± 0.3	3.7 ± 0.4	9.8 ± 0.7	16.5 ± 0.8	13.9 ± 0.8	24.0 ± 0.8	24.0 ± 0.6
	TS	66.3 ± 0.6	27.8 ± 0.4	2.4 ± 0.6	6.4 ± 0.2	10.5 ± 0.7	11.1 ± 0.4	22.0 ± 0.9	20.9 ± 0.5
	SDCLR	30.1 ± 0.3	10.0 ± 0.3	1.7 ± 0.3	2.2 ± 0.2	10.1 ± 0.7	4.4 ± 0.4	9.8 ± 0.3	9.7 ± 0.4
	BRIDGE (ours)	74.8 ± 0.4	**41.0 ± 0.4**	**4.8 ± 0.5**	**10.0 ± 0.2**	**24.2 ± 0.6**	**16.2 ± 0.7**	**29.7 ± 0.3**	**28.7 ± 0.4**
	BRIDGE+TS (ours)	**77.6 ± 0.7**	35.3 ± 0.5	3.1 ± 0.3	6.6 ± 0.4	16.3 ± 0.6	12.5 ± 1.0	26.7 ± 0.1	25.5 ± 0.5
	BRIDGE+SDCLR (ours)	34.4 ± 1.2	11.4 ± 0.6	1.2 ± 0.2	2.2 ± 0.2	8.8 ± 1.5	5.0 ± 0.2	11.7 ± 1.1	10.7 ± 0.7
CIFAR-100-LT	SimCLR	59.3 ± 0.8	30.1 ± 0.9	3.0 ± 0.3	7.4 ± 0.4	13.7 ± 0.2	10.8 ± 0.4	21.8 ± 1.1	20.9 ± 0.6
	TS	53.1 ± 0.6	24.6 ± 0.4	2.4 ± 0.4	5.5 ± 0.3	9.3 ± 0.4	8.3 ± 0.2	18.7 ± 1.0	17.4 ± 0.5
	SDCLR	25.7 ± 1.2	8.5 ± 0.6	0.9 ± 0.4	1.9 ± 0.2	7.3 ± 0.3	3.9 ± 0.3	8.3 ± 1.1	8.1 ± 0.6
	BRIDGE (ours)	**69.3 ± 0.5**	**39.5 ± 0.1**	**4.8 ± 0.6**	**8.9 ± 0.1**	**21.9 ± 0.1**	**15.5 ± 0.1**	**29.1 ± 0.7**	**27.0 ± 0.3**
	BRIDGE+TS (ours)	64.3 ± 0.4	36.6 ± 0.4	2.9 ± 0.3	6.3 ± 0.1	16.4 ± 0.5	12.3 ± 0.0	28.2 ± 0.6	23.9 ± 0.3
	BRIDGE+SDCLR (ours)	31.9 ± 0.4	10.4 ± 0.5	0.8 ± 0.1	2.1 ± 0.5	7.5 ± 1.6	4.5 ± 0.1	9.1 ± 0.7	9.5 ± 0.6
PASS-10k	SimCLR	67.2 ± 0.2	**37.0 ± 0.3**	5.4 ± 0.3	10.3 ± 0.5	30.4 ± 0.0	16.7 ± 0.5	38.3 ± 0.3	29.3 ± 0.3
	TS	65.2 ± 0.7	34.6 ± 0.3	4.4 ± 0.2	7.9 ± 0.2	24.5 ± 0.1	14.4 ± 0.4	39.3 ± 0.4	27.2 ± 0.3
	SDCLR	32.4 ± 1.5	11.5 ± 0.8	1.3 ± 0.2	2.0 ± 0.1	8.6 ± 0.7	5.2 ± 0.2	11.1 ± 0.8	10.3 ± 0.6
	BRIDGE (ours)	**67.5 ± 0.3**	36.9 ± 0.3	**5.8 ± 0.5**	**10.7 ± 0.4**	**33.7 ± 0.2**	**17.8 ± 0.6**	**41.4 ± 0.8**	**30.6 ± 0.4**
	BRIDGE+TS (ours)	58.8 ± 0.5	27.9 ± 0.5	3.0 ± 0.2	6.6 ± 0.1	23.0 ± 0.3	13.4 ± 0.4	38.8 ± 0.8	24.5 ± 0.4
	BRIDGE+SDCLR (ours)	37.1 ± 2.0	13.6 ± 1.0	1.6 ± 0.2	2.4 ± 0.2	10.1 ± 0.5	6.3 ± 0.3	13.1 ± 0.6	12.0 ± 0.7
DiffusionDB-10k	SimCLR	67.3 ± 0.1	36.2 ± 0.4	4.3 ± 0.2	8.8 ± 0.1	26.8 ± 0.4	17.3 ± 0.1	36.7 ± 0.8	28.2 ± 0.3
	TS	64.7 ± 0.4	33.6 ± 0.3	3.1 ± 0.5	7.3 ± 0.3	23.2 ± 0.4	17.0 ± 0.0	37.8 ± 0.6	26.7 ± 0.4
	SDCLR	32.6 ± 1.2	10.0 ± 0.3	1.3 ± 0.1	2.2 ± 0.2	6.1 ± 0.7	4.8 ± 0.4	10.6 ± 0.5	9.7 ± 0.5
	BRIDGE (ours)	**68.3 ± 0.2**	**37.4 ± 0.4**	**5.4 ± 0.3**	**9.8 ± 0.1**	**30.4 ± 0.2**	**18.6 ± 0.2**	**39.4 ± 1.0**	**29.9 ± 0.3**
	BRIDGE+TS (ours)	57.5 ± 0.8	26.6 ± 0.3	3.0 ± 0.4	6.5 ± 0.1	21.7 ± 0.8	15.5 ± 0.4	37.0 ± 0.4	24.0 ± 0.5
	BRIDGE+SDCLR (ours)	35.2 ± 0.7	11.1 ± 0.3	1.0 ± 0.1	2.2 ± 0.1	8.1 ± 1.0	5.0 ± 0.3	11.3 ± 0.7	10.6 ± 0.5

Table 11: Pretraining objective ablation on the default ResNet-50 backbone, full linear-probe results. DINO is excluded here because its reference recipe is defined for ViT backbones; the DINO comparison is reported with ViT-S in Table 3.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
SimCLR (default)	**76.0 ± 0.3**	**49.1 ± 0.2**	**15.2 ± 1.2**	**22.4 ± 0.4**	**36.8 ± 1.9**	**42.9 ± 1.0**	**53.8 ± 0.1**	**42.3 ± 0.7**
MoCo	50.5 ± 1.0	20.6 ± 2.7	3.0 ± 0.2	5.7 ± 0.5	11.5 ± 3.0	15.3 ± 0.3	28.2 ± 0.7	19.3 ± 1.2

Table 12: Pretraining objective ablation on the default ResNet-50 backbone, full kNN results. DINO is excluded here because its reference recipe is defined for ViT backbones; the DINO comparison is reported with ViT-S in Table 3.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
SimCLR (default)	**69.0 ± 0.5**	**37.8 ± 0.3**	**6.7 ± 0.3**	**11.7 ± 0.2**	**36.5 ± 0.4**	**25.3 ± 1.0**	**48.1 ± 0.6**	**33.6 ± 0.5**
MoCo	58.0 ± 0.4	26.7 ± 0.5	3.0 ± 0.2	5.4 ± 0.2	16.3 ± 0.9	12.2 ± 0.6	25.1 ± 0.5	21.0 ± 0.5

Table 13: Augmentation mechanism ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Stable Diffusion 3 (default)	**75.0 ± 0.3**	**48.5 ± 0.2**	**16.3 ± 0.8**	**20.7 ± 0.4**	35.7 ± 0.9	**41.8 ± 0.8**	**52.4 ± 0.7**	**41.5 ± 0.6**
FLUX	70.7 ± 0.4	42.1 ± 1.4	10.3 ± 1.3	17.5 ± 0.9	29.4 ± 4.6	31.5 ± 2.9	43.7 ± 1.4	35.0 ± 1.9
No-generation re-population	74.8 ± 0.4	47.8 ± 0.7	13.1 ± 0.1	20.5 ± 0.5	**40.3 ± 6.1**	38.1 ± 1.5	50.3 ± 0.4	40.7 ± 1.4

Table 14: Augmentation mechanism ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Stable Diffusion 3 (default)	68.6 ± 0.1	37.2 ± 0.2	6.4 ± 0.6	**11.5 ± 0.2**	**35.2 ± 0.3**	**24.3 ± 0.6**	**46.8 ± 0.7**	**32.9 ± 0.4**
FLUX	65.8 ± 1.0	35.4 ± 0.8	5.7 ± 0.7	9.4 ± 0.9	27.7 ± 1.0	18.8 ± 1.3	38.7 ± 1.2	28.8 ± 1.0
No-generation re-population	**68.8 ± 0.2**	**37.7 ± 0.3**	**6.6 ± 0.5**	11.4 ± 0.5	32.7 ± 0.1	23.6 ± 0.6	44.4 ± 0.4	32.2 ± 0.4

Table 15: Selection rule ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Mode-window q1-99 + diversity	75.6 ± 0.1	49.2 ± 0.2	**16.9 ± 0.7**	21.9 ± 0.4	34.6 ± 2.9	43.5 ± 0.7	52.8 ± 0.1	42.1 ± 0.7
Mode-window q50-99 + diversity	75.5 ± 0.1	**49.3 ± 0.4**	15.2 ± 0.3	**22.4 ± 0.6**	35.5 ± 1.1	**43.8 ± 3.1**	53.4 ± 0.9	42.2 ± 0.9
Mode-window q75-99 + diversity (default)	**76.0 ± 0.3**	49.1 ± 0.2	15.2 ± 1.2	22.4 ± 0.4	**36.8 ± 1.9**	42.9 ± 1.0	**53.8 ± 0.1**	**42.3 ± 0.7**
Top-tail	70.8 ± 0.5	43.1 ± 1.0	10.0 ± 1.7	15.9 ± 1.3	27.7 ± 1.0	30.5 ± 1.2	42.2 ± 1.4	34.3 ± 1.2
Uniform	71.5 ± 0.2	43.8 ± 0.7	9.3 ± 0.9	17.5 ± 1.2	32.9 ± 0.4	31.7 ± 3.9	42.0 ± 0.9	35.5 ± 1.2

Table 16: Selection rule ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Mode-window q1-99 + diversity	**69.2 ± 0.3**	37.7 ± 0.2	6.6 ± 0.2	11.4 ± 0.2	35.9 ± 0.2	25.0 ± 0.5	48.3 ± 0.3	33.4 ± 0.3
Mode-window q50-99 + diversity	68.7 ± 0.1	**38.0 ± 0.2**	**7.3 ± 0.2**	11.3 ± 0.2	**36.5 ± 0.1**	24.8 ± 0.8	**48.5 ± 0.4**	33.6 ± 0.3
Mode-window q75-99 + diversity (default)	69.0 ± 0.5	37.8 ± 0.3	6.7 ± 0.3	**11.7 ± 0.2**	36.5 ± 0.4	**25.3 ± 1.0**	48.1 ± 0.6	**33.6 ± 0.5**
Top-tail	65.7 ± 0.8	34.4 ± 0.2	4.1 ± 0.4	9.6 ± 0.2	23.9 ± 0.9	18.1 ± 0.8	35.4 ± 1.2	27.3 ± 0.6
Uniform	66.1 ± 1.0	35.7 ± 0.3	4.9 ± 0.1	10.2 ± 0.7	25.3 ± 0.4	19.0 ± 0.8	37.4 ± 0.6	28.4 ± 0.6

Table 17: Cycle count ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
2 cycles	68.9 ± 0.6	41.6 ± 1.3	10.6 ± 1.2	16.4 ± 0.4	30.7 ± 4.5	33.2 ± 0.9	43.3 ± 0.6	35.0 ± 1.4
5 cycles (default)	**76.0 ± 0.3**	**49.1 ± 0.2**	**15.2 ± 1.2**	**22.4 ± 0.4**	**36.8 ± 1.9**	**42.9 ± 1.0**	**53.8 ± 0.1**	**42.3 ± 0.7**
10 cycles	70.5 ± 1.2	43.8 ± 0.6	9.2 ± 0.9	17.2 ± 1.6	24.8 ± 5.2	31.3 ± 1.7	41.9 ± 0.6	34.1 ± 1.7

Table 18: Cycle count ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
2 cycles	65.3 ± 0.4	34.2 ± 0.5	5.0 ± 0.7	10.0 ± 0.2	26.1 ± 0.8	18.7 ± 0.6	35.9 ± 0.3	27.9 ± 0.5
5 cycles (default)	**69.0 ± 0.5**	**37.8 ± 0.3**	**6.7 ± 0.3**	**11.7 ± 0.2**	**36.5 ± 0.4**	**25.3 ± 1.0**	**48.1 ± 0.6**	**33.6 ± 0.5**
10 cycles	66.4 ± 0.5	35.4 ± 0.3	4.8 ± 0.3	10.4 ± 0.6	25.2 ± 0.8	18.6 ± 0.9	36.0 ± 0.7	28.1 ± 0.6

Table 19: Backbone architecture ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ResNet-18	68.8 ± 0.1	41.7 ± 0.2	13.5 ± 0.3	19.7 ± 0.6	38.7 ± 3.4	38.4 ± 2.6	50.2 ± 0.2	38.7 ± 1.1
ResNet-50 (default)	**76.0 ± 0.3**	**49.1 ± 0.2**	**15.2 ± 1.2**	**22.4 ± 0.4**	36.8 ± 1.9	**42.9 ± 1.0**	**53.8 ± 0.1**	**42.3 ± 0.7**
ViT-S	62.7 ± 0.4	37.6 ± 0.6	8.8 ± 0.5	12.2 ± 0.2	35.1 ± 4.2	30.2 ± 1.6	42.5 ± 1.2	32.7 ± 1.3
ViT-B	66.0 ± 2.3	40.6 ± 2.3	8.3 ± 0.5	14.3 ± 0.8	**38.8 ± 2.6**	31.3 ± 2.1	44.5 ± 1.7	34.8 ± 1.8

Table 20: Backbone architecture ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ResNet-18	66.3 ± 0.2	35.7 ± 0.3	5.9 ± 0.2	11.1 ± 0.4	33.5 ± 0.1	23.3 ± 0.5	44.5 ± 0.7	31.5 ± 0.4
ResNet-50 (default)	69.0 ± 0.5	37.8 ± 0.3	**6.7 ± 0.3**	**11.7 ± 0.2**	36.5 ± 0.4	**25.3 ± 1.0**	**48.1 ± 0.6**	**33.6 ± 0.5**
ViT-S	67.3 ± 0.4	37.5 ± 0.3	5.3 ± 0.2	7.6 ± 0.1	36.2 ± 0.7	20.1 ± 1.0	42.1 ± 1.0	30.9 ± 0.5
ViT-B	**69.3 ± 0.8**	**40.1 ± 0.8**	6.0 ± 0.2	8.3 ± 0.2	**39.8 ± 0.8**	21.9 ± 1.4	45.1 ± 0.7	32.9 ± 0.7

Table 21: Selector robustness on ImageNet-100-LT. Each row changes one component of the acquisition rule – candidate pool multiplier, upper cutoff, neighbourhood size, distance, or selection strategy – and holds the rest of the loop fixed. Mean $\pm$ standard deviation over three seeds. Cosine scoring rescales the normalized L_2 score by a positive constant and therefore selects the identical anchor set; that row is a null control, and its distance from the normalized L_2 row measures the end-to-end run-to-run spread of the pipeline. Linear-probe accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
Candidate pool $\alpha = 1$	6.46 ± 0.57	11.37 ± 1.59	23.53 ± 3.00	21.92 ± 2.00	61.94 ± 0.94	32.73 ± 0.21	33.92 ± 1.68
Candidate pool $\alpha = 2$	6.77 ± 0.63	11.59 ± 0.40	21.79 ± 3.22	23.19 ± 3.93	62.90 ± 1.06	34.48 ± 0.11	35.69 ± 0.87
Candidate pool $\alpha = 4$ (default)	6.31 ± 0.10	11.17 ± 3.78	23.53 ± 3.27	24.18 ± 2.16	62.11 ± 0.19	33.39 ± 0.64	34.18 ± 0.47
Candidate pool $\alpha = 8$	6.41 ± 0.28	11.16 ± 0.78	23.09 ± 4.35	24.82 ± 3.29	62.60 ± 1.79	33.15 ± 1.01	35.40 ± 0.87
Upper cutoff $q = 0.95$	6.79 ± 0.65	11.44 ± 0.85	19.83 ± 2.47	25.27 ± 2.32	62.57 ± 0.87	33.83 ± 0.53	35.13 ± 1.03
Upper cutoff $q = 0.99$ (default)	6.59 ± 0.17	11.76 ± 1.54	20.26 ± 1.73	24.00 ± 2.20	63.11 ± 0.75	33.43 ± 1.02	35.85 ± 1.15
Upper cutoff $q = 0.995$	6.78 ± 0.66	10.40 ± 2.49	22.88 ± 4.58	22.55 ± 3.09	62.76 ± 0.50	33.87 ± 0.38	35.32 ± 0.64
Upper cutoff $q = 1.0$	6.31 ± 0.45	10.94 ± 1.34	22.44 ± 1.89	23.37 ± 4.00	62.85 ± 1.04	33.91 ± 0.87	35.51 ± 1.59
Neighbourhood $k = 10$	6.26 ± 0.29	10.44 ± 1.43	22.22 ± 0.65	24.00 ± 1.23	62.66 ± 0.60	33.44 ± 0.58	35.14 ± 1.21
Neighbourhood $k = 25$	6.44 ± 0.76	11.13 ± 1.19	19.39 ± 3.60	23.37 ± 1.18	63.18 ± 0.56	33.40 ± 1.21	33.49 ± 0.43
Neighbourhood $k = 50$	6.34 ± 0.56	10.01 ± 1.32	20.48 ± 1.00	22.64 ± 3.24	61.28 ± 2.19	32.64 ± 0.29	34.23 ± 0.77
Neighbourhood $k = 200$	6.50 ± 0.46	10.58 ± 2.83	20.26 ± 1.13	22.46 ± 1.23	61.24 ± 1.08	32.87 ± 0.76	34.42 ± 1.23
Distance: normalized L_2 (default)	6.59 ± 0.17	11.79 ± 1.34	20.48 ± 2.64	24.46 ± 2.68	61.69 ± 0.67	33.88 ± 1.05	35.24 ± 1.10
Distance: cosine (null control)	6.55 ± 0.15	10.50 ± 1.10	20.04 ± 2.64	23.91 ± 1.70	62.22 ± 1.29	33.23 ± 0.81	34.98 ± 0.51
Distance: raw L_2	6.92 ± 0.78	10.45 ± 3.00	18.95 ± 5.23	25.72 ± 2.31	62.81 ± 0.80	33.20 ± 0.89	35.08 ± 2.48
Strategy: mode window (default)	6.55 ± 0.15	10.18 ± 1.30	22.44 ± 4.59	23.28 ± 2.04	60.95 ± 3.01	33.05 ± 0.21	34.53 ± 0.92
Strategy: sparse band with FPS	6.60 ± 0.62	11.21 ± 1.26	21.13 ± 2.64	22.46 ± 0.16	63.34 ± 0.27	34.29 ± 0.30	34.30 ± 1.26
Strategy: sparse band, random	6.46 ± 0.48	11.06 ± 0.70	21.35 ± 4.82	22.92 ± 0.87	62.26 ± 0.89	32.33 ± 0.96	34.46 ± 1.08
Strategy: inverse-cluster	6.44 ± 0.75	10.66 ± 1.06	19.17 ± 2.47	21.65 ± 2.28	62.70 ± 0.45	33.50 ± 0.95	34.69 ± 1.10
Strategy: densest window	6.64 ± 0.46	11.90 ± 0.49	20.04 ± 1.00	24.28 ± 1.64	63.50 ± 0.60	33.80 ± 0.93	35.20 ± 0.80
Strategy: top tail, no diversity	6.39 ± 0.43	10.48 ± 1.03	19.83 ± 2.00	23.82 ± 1.50	62.86 ± 0.92	33.14 ± 1.02	34.66 ± 0.41

Table 22: Selector robustness on ImageNet-100-LT. Each row changes one component of the acquisition rule – candidate pool multiplier, upper cutoff, neighbourhood size, distance, or selection strategy – and holds the rest of the loop fixed. Mean $\pm$ standard deviation over three seeds. Cosine scoring rescales the normalized L_2 score by a positive constant and therefore selects the identical anchor set; that row is a null control, and its distance from the normalized L_2 row measures the end-to-end run-to-run spread of the pipeline. kNN accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
Candidate pool $\alpha = 1$	5.02 ± 0.14	8.62 ± 0.26	19.22 ± 0.71	14.92 ± 0.99	62.40 ± 0.12	31.47 ± 0.25	33.47 ± 1.05
Candidate pool $\alpha = 2$	5.08 ± 0.15	8.36 ± 0.22	19.07 ± 0.42	15.17 ± 0.27	62.81 ± 0.58	31.72 ± 0.27	33.19 ± 0.61
Candidate pool $\alpha = 4$ (default)	4.92 ± 0.15	8.48 ± 0.72	19.04 ± 0.03	14.60 ± 0.38	62.35 ± 0.54	31.84 ± 0.37	32.59 ± 0.49
Candidate pool $\alpha = 8$	4.85 ± 0.17	8.53 ± 0.41	19.17 ± 0.90	14.82 ± 0.62	62.15 ± 0.07	31.56 ± 0.23	33.58 ± 1.03
Upper cutoff $q = 0.95$	4.99 ± 0.11	8.57 ± 0.51	19.09 ± 0.51	14.68 ± 0.28	62.84 ± 0.45	31.88 ± 0.04	32.56 ± 0.45
Upper cutoff $q = 0.99$ (default)	4.95 ± 0.17	8.53 ± 0.69	19.40 ± 0.45	14.85 ± 0.49	62.29 ± 0.50	31.87 ± 0.16	32.44 ± 0.48
Upper cutoff $q = 0.995$	5.01 ± 0.07	8.34 ± 0.74	19.58 ± 0.83	14.87 ± 0.48	63.05 ± 0.47	32.06 ± 0.11	33.01 ± 0.34
Upper cutoff $q = 1.0$	4.99 ± 0.08	8.15 ± 0.61	18.52 ± 0.98	14.45 ± 0.19	62.84 ± 0.32	31.69 ± 0.30	32.99 ± 0.83
Neighbourhood $k = 10$	5.04 ± 0.09	8.99 ± 1.21	18.86 ± 0.35	14.90 ± 0.57	62.73 ± 0.39	31.69 ± 0.17	33.14 ± 0.40
Neighbourhood $k = 25$	4.97 ± 0.14	8.59 ± 0.21	19.05 ± 0.29	15.04 ± 0.85	62.68 ± 0.66	31.82 ± 0.16	33.56 ± 0.97
Neighbourhood $k = 50$	5.06 ± 0.17	8.62 ± 0.77	19.26 ± 0.67	14.92 ± 0.03	62.63 ± 0.40	31.71 ± 0.59	31.88 ± 0.57
Neighbourhood $k = 200$	5.17 ± 0.15	8.65 ± 0.59	19.22 ± 1.02	15.17 ± 0.10	62.80 ± 0.31	31.90 ± 0.13	33.91 ± 0.54
Distance: normalized L_2 (default)	4.97 ± 0.18	8.55 ± 0.67	19.42 ± 0.49	14.85 ± 0.50	62.27 ± 0.52	31.86 ± 0.16	32.60 ± 0.96
Distance: cosine (null control)	4.90 ± 0.13	8.52 ± 0.69	19.35 ± 0.54	14.75 ± 0.44	62.31 ± 0.53	31.78 ± 0.28	32.44 ± 0.83
Distance: raw L_2	4.85 ± 0.10	7.83 ± 0.39	19.13 ± 0.88	14.63 ± 0.34	62.81 ± 0.52	31.71 ± 0.04	32.28 ± 0.65
Strategy: mode window (default)	4.90 ± 0.12	8.52 ± 0.69	19.34 ± 0.54	14.75 ± 0.45	62.31 ± 0.52	31.78 ± 0.28	32.73 ± 0.75
Strategy: sparse band with FPS	5.19 ± 0.04	8.12 ± 0.39	19.53 ± 0.49	14.95 ± 0.75	62.43 ± 0.17	31.62 ± 0.32	33.38 ± 1.46
Strategy: sparse band, random	5.00 ± 0.13	8.70 ± 0.06	19.36 ± 0.88	14.95 ± 0.61	62.63 ± 0.19	31.64 ± 0.62	33.04 ± 0.82
Strategy: inverse-cluster	5.05 ± 0.02	8.83 ± 1.05	19.40 ± 0.81	15.06 ± 0.46	62.60 ± 0.50	31.99 ± 0.37	33.75 ± 0.44
Strategy: densest window	4.88 ± 0.10	8.44 ± 0.54	19.34 ± 0.59	14.95 ± 0.53	62.46 ± 0.12	31.62 ± 0.30	33.08 ± 0.64
Strategy: top tail, no diversity	4.90 ± 0.12	8.38 ± 0.29	19.11 ± 0.64	15.22 ± 0.03	62.05 ± 0.77	31.77 ± 0.13	32.41 ± 0.25

Table 23: Repair utility by sparsity band on ImageNet-100-LT. Each row spends the identical acquisition budget on a different quantile range of the kNN score distribution, holding every other part of the loop fixed. Mean $\pm$ standard deviation over three seeds. kNN accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
q0–25 (densest)	4.97 ± 0.07	8.33 ± 0.72	20.00 ± 0.65	14.48 ± 0.27	62.04 ± 0.84	31.45 ± 0.05	31.88 ± 0.91
q25–50	5.16 ± 0.14	8.52 ± 0.37	19.58 ± 0.49	14.82 ± 0.73	62.95 ± 0.79	31.54 ± 0.09	32.88 ± 0.78
q50–75	5.06 ± 0.08	8.51 ± 0.44	19.63 ± 0.75	14.46 ± 0.81	62.41 ± 0.61	31.47 ± 0.56	32.61 ± 0.55
q75–85	4.97 ± 0.08	8.78 ± 0.45	19.54 ± 0.14	15.39 ± 0.20	62.52 ± 0.56	31.31 ± 0.43	32.23 ± 1.17
q85–90	5.04 ± 0.09	9.04 ± 0.48	19.61 ± 0.46	14.75 ± 0.65	62.57 ± 0.27	31.74 ± 0.34	32.82 ± 0.96
q90–95	4.92 ± 0.21	8.67 ± 0.75	19.60 ± 0.53	14.96 ± 0.46	62.40 ± 0.59	31.56 ± 0.54	32.48 ± 0.56
q95–97	4.88 ± 0.35	8.51 ± 1.09	19.40 ± 0.72	14.97 ± 0.49	62.32 ± 0.94	31.55 ± 0.43	31.63 ± 0.32
q97–99	5.02 ± 0.08	8.59 ± 0.43	19.63 ± 0.67	14.89 ± 0.56	62.64 ± 0.56	31.75 ± 0.61	32.59 ± 0.96
q99–100 (extreme tail)	5.17 ± 0.16	8.16 ± 0.23	19.10 ± 0.60	15.00 ± 0.53	62.81 ± 0.26	32.12 ± 0.43	32.15 ± 1.40

Table 24: Paired full-schedule ViT-S transfer on ImageNet-100-LT, with every downstream dataset reported separately. Both arms of each pair start from the same source checkpoint and receive the same number of post-branch optimizer updates. Five objectives span contrastive (SimCLR, MoCo v3), self-distillation (DINO, DINOv2) and masked-reconstruction (MAE) pretraining. SimCLR, MoCo v3 and DINO run five cycles of 100 epochs; DINOv2 and MAE run five of 80, since DINOv2 carries multi-crop at batch 16 and does not finish otherwise. Mean $\pm$ standard deviation over three seeds. Linear-probe accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
SimCLR, SSL baseline	7.15 ± 0.19	10.92 ± 0.40	32.90 ± 2.72	25.00 ± 1.25	60.46 ± 0.46	34.96 ± 0.20	41.95 ± 1.43
SimCLR, with BRIDGE	8.18 ± 0.42	11.82 ± 0.43	36.60 ± 2.85	29.80 ± 2.98	60.85 ± 0.35	36.33 ± 0.14	43.93 ± 1.09
MoCo v3, SSL baseline	7.57 ± 0.23	11.05 ± 1.31	30.28 ± 1.36	29.26 ± 3.13	59.96 ± 0.39	34.47 ± 0.01	44.79 ± 1.17
MoCo v3, with BRIDGE	9.55 ± 0.34	13.42 ± 0.98	36.82 ± 6.07	31.25 ± 4.00	62.36 ± 0.46	36.30 ± 0.13	47.55 ± 0.86
DINO, SSL baseline	8.10 ± 0.25	11.40 ± 1.03	38.13 ± 5.24	27.54 ± 2.34	58.93 ± 0.94	33.56 ± 1.75	44.63 ± 1.47
DINO, with BRIDGE	9.49 ± 0.88	13.28 ± 0.58	37.25 ± 2.61	34.60 ± 3.62	61.30 ± 0.07	35.23 ± 0.31	46.78 ± 0.77
DINOv2, SSL baseline	2.13 ± 0.35	3.32 ± 0.47	14.16 ± 2.64	9.15 ± 1.39	33.47 ± 0.87	11.62 ± 1.82	18.68 ± 1.21
DINOv2, with BRIDGE	2.06 ± 0.16	3.25 ± 0.15	12.42 ± 3.46	7.25 ± 0.87	33.21 ± 1.49	10.71 ± 1.44	19.35 ± 1.34
MAE, SSL baseline	1.54 ± 0.19	2.44 ± 1.00	10.46 ± 5.88	10.05 ± 3.03	37.69 ± 3.73	13.20 ± 1.94	19.93 ± 2.13
MAE, with BRIDGE	1.56 ± 0.43	3.18 ± 0.70	8.50 ± 2.26	8.61 ± 2.20	41.46 ± 2.20	15.00 ± 2.40	22.19 ± 2.35

I ALGORITHM

Algorithm 1 BRIDGE training loop with mode-window selection

```
Initialize source set D^(0), encoder f_θ(0), frozen generator g_φ, cycle budgets {E_c}_{c=0}^{C−1}
for c = 0, ..., C − 1 do
    Train or continue training f_θ(c) on D^(c) for E_c epochs
    Compute embeddings Z^(c) = {f_θ(c)(x_i) : x_i ∈ D^(c)}
    Compute mean kNN scores {d_i^(c)}_{i=1}^{N^(c)} from Z^(c)
    Restrict {d_i^(c)} to the 75th to 99th percentile range and build a histogram
    Let m^(c) be the center of the modal histogram bin inside that upper-tail band
    Form a candidate set C^(c) of size 4B whose scores are closest to m^(c)
    Select S^(c) ⊆ C^(c) with |S^(c)| = B by greedy farthest-point sampling in embedding space
    if c < C − 1 then
        Form D̃^(c) = {g_φ(x_i, ε_j) | i ∈ S^(c), j ∈ [G]}
        Update D^(c+1) = D^(c) ∪ D̃^(c)
    end if
end for
Return the final encoder and repaired source set
```

Table 25: Paired full-schedule ViT-S transfer on ImageNet-100-LT, with every downstream dataset reported separately. Both arms of each pair start from the same source checkpoint and receive the same number of post-branch optimizer updates. Five objectives span contrastive (SimCLR, MoCo v3), self-distillation (DINO, DINOv2) and masked-reconstruction (MAE) pretraining. SimCLR, MoCo v3 and DINO run five cycles of 100 epochs; DINOv2 and MAE run five of 80, since DINOv2 carries multi-crop at batch 16 and does not finish otherwise. Mean $\pm$ standard deviation over three seeds. kNN accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
SimCLR, SSL baseline	6.26 ± 0.17	7.09 ± 0.17	36.53 ± 0.89	19.16 ± 0.64	66.05 ± 0.48	36.72 ± 0.36	41.21 ± 0.72
SimCLR, with BRIDGE	6.52 ± 0.22	7.56 ± 0.23	38.29 ± 0.83	19.93 ± 0.53	65.70 ± 0.85	36.46 ± 0.33	42.91 ± 0.78
MoCo v3, SSL baseline	6.65 ± 0.38	7.71 ± 0.16	38.91 ± 1.03	23.56 ± 1.12	69.29 ± 0.25	39.98 ± 0.44	45.14 ± 0.46
MoCo v3, with BRIDGE	6.74 ± 0.26	8.21 ± 0.18	40.51 ± 1.45	24.91 ± 0.45	65.41 ± 0.21	36.24 ± 0.46	47.71 ± 1.53
DINO, SSL baseline	6.05 ± 0.27	7.73 ± 0.52	38.52 ± 1.19	19.26 ± 0.16	62.69 ± 2.42	33.98 ± 2.80	45.28 ± 0.98
DINO, with BRIDGE	6.39 ± 0.22	8.16 ± 0.26	40.72 ± 0.31	21.79 ± 1.15	64.38 ± 3.51	35.59 ± 2.51	47.14 ± 0.29
DINOv2, SSL baseline	2.28 ± 0.37	3.49 ± 0.15	18.02 ± 1.15	5.53 ± 0.36	33.90 ± 1.34	13.68 ± 0.45	14.79 ± 1.66
DINOv2, with BRIDGE	2.55 ± 0.05	3.46 ± 0.46	18.43 ± 1.66	6.21 ± 0.35	35.33 ± 1.43	14.53 ± 0.70	14.85 ± 0.92
MAE, SSL baseline	1.00 ± 0.14	2.24 ± 0.29	5.30 ± 0.52	4.58 ± 0.17	31.34 ± 1.65	10.96 ± 0.44	11.75 ± 0.38
MAE, with BRIDGE	1.38 ± 0.37	2.50 ± 0.70	7.54 ± 2.13	5.35 ± 0.76	35.58 ± 7.14	13.44 ± 3.46	15.46 ± 2.68

Table 26: Matched feedback and repair controls on ImageNet-100-LT. All methods start from paired source checkpoints and receive three 60-epoch stages with the same update cap. Generation methods add 7,500 images in total. Each entry is mean $\pm$ standard deviation over three seeds. Linear-probe accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
Adaptive BRIDGE	12.68 ± 0.88	17.14 ± 0.92	27.89 ± 4.99	32.43 ± 2.00	68.60 ± 0.89	40.28 ± 0.82	42.34 ± 0.69
Frozen first-cycle selector	12.70 ± 0.78	16.89 ± 0.98	25.93 ± 4.73	30.16 ± 1.63	68.72 ± 0.65	40.95 ± 1.15	43.64 ± 0.45
One-shot repair	12.73 ± 0.43	18.13 ± 0.46	28.98 ± 2.72	32.70 ± 0.78	68.72 ± 1.42	40.71 ± 1.73	43.66 ± 0.76
No repair	12.33 ± 0.53	16.49 ± 1.01	28.76 ± 1.96	32.25 ± 2.99	67.76 ± 0.45	39.83 ± 0.71	42.56 ± 0.74
Mode-window conventional augmentation	12.04 ± 0.18	17.01 ± 1.05	22.66 ± 4.35	30.71 ± 3.20	68.40 ± 0.53	40.13 ± 0.42	41.28 ± 0.46
Uniform acquisition	12.83 ± 0.42	17.73 ± 0.80	27.23 ± 6.07	31.16 ± 4.01	68.86 ± 0.66	41.09 ± 1.27	43.10 ± 0.78
Top-tail acquisition	12.12 ± 0.35	17.18 ± 0.90	27.89 ± 2.64	32.07 ± 2.05	69.16 ± 0.94	41.17 ± 0.93	43.27 ± 0.77
AIDE-style VLM acquisition and text-to-image	12.28 ± 0.15	17.86 ± 0.55	28.76 ± 0.65	33.42 ± 0.94	69.78 ± 0.37	41.39 ± 1.03	43.36 ± 0.08
Mode-window caption and text-to-image	12.42 ± 1.13	17.51 ± 1.47	25.05 ± 3.72	26.09 ± 1.44	64.25 ± 0.85	35.82 ± 0.47	43.64 ± 0.79
Mode-window captioned image-to-image	12.66 ± 0.56	17.59 ± 0.30	25.27 ± 2.95	26.99 ± 4.79	65.31 ± 4.85	37.49 ± 5.22	43.13 ± 0.61

Table 27: Matched feedback and repair controls on ImageNet-100-LT. All methods start from paired source checkpoints and receive three 60-epoch stages with the same update cap. Generation methods add 7,500 images in total. Each entry is mean $\pm$ standard deviation over three seeds. kNN accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
Adaptive BRIDGE	6.12 ± 0.11	10.20 ± 0.40	23.41 ± 0.48	17.13 ± 0.58	65.60 ± 0.76	34.65 ± 0.34	35.11 ± 0.28
Frozen first-cycle selector	6.02 ± 0.05	10.20 ± 0.13	23.60 ± 0.36	16.69 ± 0.53	65.39 ± 0.78	35.07 ± 0.30	35.38 ± 0.26
One-shot repair	6.03 ± 0.14	10.19 ± 0.26	23.73 ± 0.96	16.93 ± 0.67	65.87 ± 0.36	35.31 ± 0.40	35.57 ± 0.53
No repair	5.77 ± 0.24	9.88 ± 0.81	23.33 ± 0.39	17.02 ± 0.14	66.10 ± 0.33	35.27 ± 0.37	35.27 ± 0.46
Mode-window conventional augmentation	5.78 ± 0.20	10.04 ± 0.66	22.74 ± 0.58	16.44 ± 0.99	65.89 ± 0.24	35.19 ± 0.05	36.17 ± 1.17
Uniform acquisition	6.16 ± 0.10	10.29 ± 0.54	23.74 ± 0.75	17.17 ± 0.87	65.60 ± 0.44	34.71 ± 0.54	35.44 ± 0.76
Top-tail acquisition	6.01 ± 0.22	10.33 ± 0.37	23.16 ± 0.48	17.06 ± 0.12	65.54 ± 0.48	34.97 ± 0.16	35.45 ± 0.50
AIDE-style VLM acquisition and text-to-image	5.96 ± 0.11	10.06 ± 0.44	23.51 ± 0.60	17.23 ± 0.15	66.35 ± 0.22	35.44 ± 0.27	34.99 ± 0.25
Mode-window caption and text-to-image	4.95 ± 0.12	8.61 ± 0.68	18.84 ± 0.31	14.26 ± 0.61	62.68 ± 0.18	31.84 ± 0.34	31.92 ± 0.38
Mode-window captioned image-to-image	5.25 ± 0.61	9.08 ± 0.53	20.01 ± 2.13	15.14 ± 1.43	63.64 ± 2.10	33.23 ± 1.99	33.20 ± 2.07